

IE437 Data-Driven Decision Making and Control
2023 Spring Semester

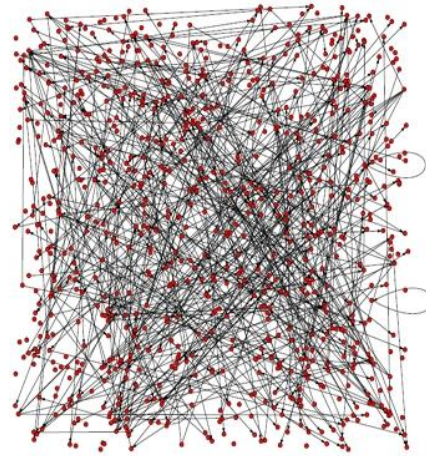
3. Bayesian Network

Jinkyoo Park
Associate Professor
Industrial and Systems Engineering Department

Bayesian Network

Motivation

Bayesian Network is System Modeling



Probability + **Statistics** + **Graph Theory**

Degree of belief and probability

- How to compare the plausibility of different statements?



G : “we can be a billionaire if we go to graduate school”

vs

S : “we can be a billionaire if we go to Samsung”



- If you believe G more than S , you can write $G \succ S$
- If you believe S more than G , you can write $G \prec S$
- If you have the same belief, you can write $G \sim S$
- Assumptions about relationships of $\succ$ and $\sim$
 - Universal comparability: either $G \succ S$, $G \prec S$ or $G \sim S$
 - Transitivity: if $G \succ S$ and $S \succ V$, then $G \succ V$
- The **degree of belief** can be represented by a real-valued function:
 - $P(G) > P(S)$ if and only if $G \succ S$
 - $P(G) = P(S)$ if and only if $G \sim S$

Properties of probabilities for Bayesian Networks

- We are going to use very simple probability theories to construct Probabilistic Graphical Model

- **Conditional probability:**

$$P(A|B) = \frac{P(B|A)}{P(B)}$$

- **Law of total probability:**

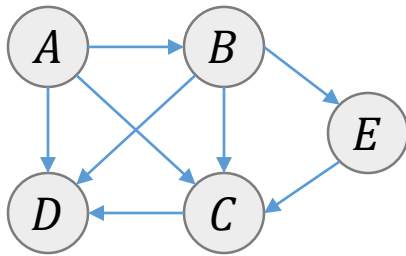
$$P(A) = \sum_{B \in \mathcal{B}} P(A|B)P(B)$$

- **Bayes' rule:**

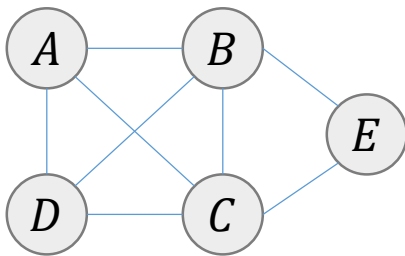
$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

Introduction to Graph Theory

- A **graph** G consists of nodes (also called vertices) and edges (also called links) between the nodes.



A **directed graph** G consists of directed edges between nodes

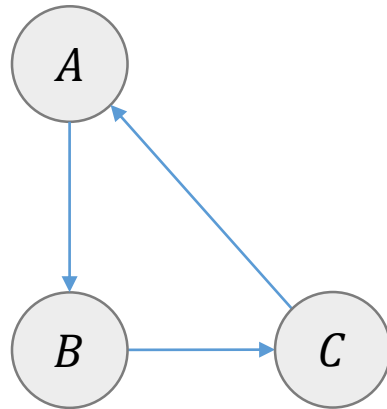


An **undirected graph** G consists of undirected edges between nodes

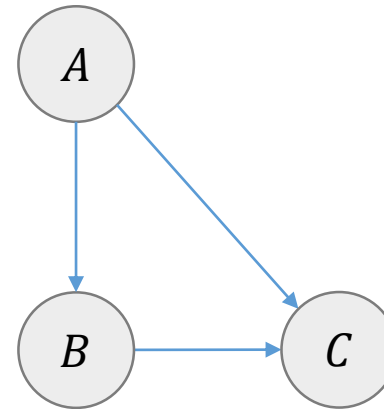
Introduction to Graph Theory

- A **Directed Acyclic Graph (DAG)** is a graph G with directed edges (arrows on each link) between nodes such that by following a path of nodes from one node to another along the direction of each edge no path will revisit a node.

Cyclic Graph



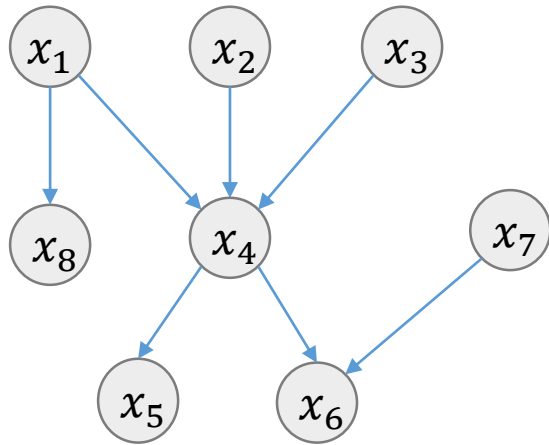
Acyclic Graph



- DAG will play a central role in modeling environments with many variables
 - will be used for the belief networks
 - can encode the direction dependence between the parent nodes and child nodes.

Introduction to Graph Theory

- A **path** $A \rightarrow B$ from node A to node B is a sequence of nodes that connects A to B
- In directed graph, the nodes A such that $A \rightarrow B$ and $B \nrightarrow A$ are the **ancestors** of B
- In directed graph, the nodes B such that $A \rightarrow B$ and $B \nrightarrow A$ are the **descendants** of A



- ✓ A path $x_1 \rightarrow x_6$ is $x_1 \rightarrow x_4 \rightarrow x_6$
- ✓ The ancestors of x_6 are $ac(x_6) = \{x_1, x_2, x_3, x_4, x_7\}$
- ✓ The descendants of x_2 are $dc(x_2) = \{x_4, x_5, x_6\}$
- ✓ The parents of x_4 are $pa(x_4) = \{x_1, x_2, x_3\}$
- ✓ The children of x_4 are $ch(x_4) = \{x_5, x_6\}$

Representations

- **Edge list**

$$E = \{(x_1, x_4), (x_1, x_8), (x_2, x_4), (x_3, x_4), (x_4, x_5), (x_4, x_6), (x_7, x_6)\}$$

- **Adjacency matrix**

$$\begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Motivation of Bayesian network

- Full joint Distribution

Example distribution

A	B	C	$P(A, B, C)$
0	0	0	0.08
0	0	1	0.15
0	1	0	0.05
0	1	1	0.10
1	0	0	0.14
1	0	1	0.18
1	1	0	0.19
1	1	1	0.11

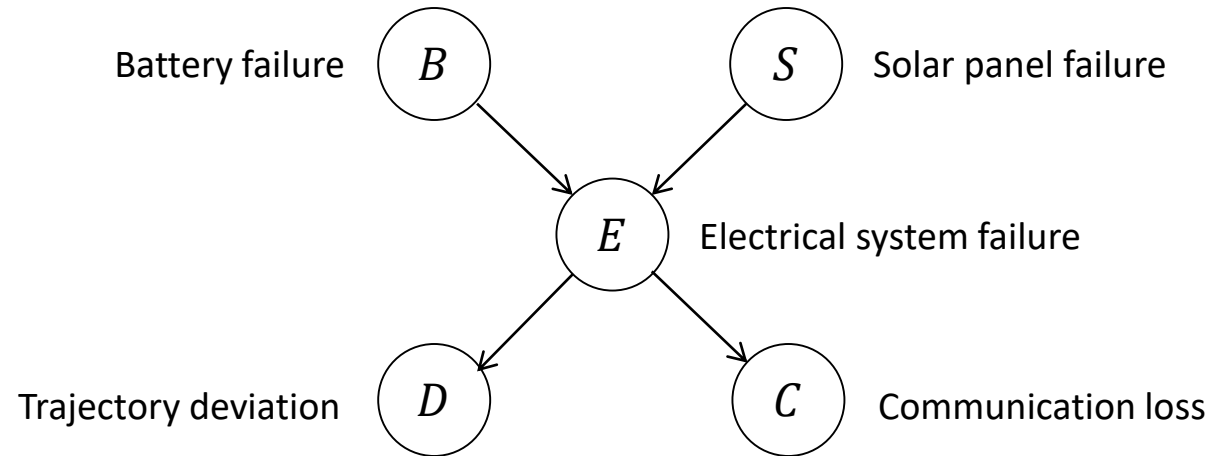
- Binary variables: A, B, C (e.g., $A = 1$ or 0)
- 2^3 entities are required to construct the table
- $2^3 - 1$ independent parameters are required to fully specify the joint probability distribution
- $2^N - 1$ parameters are required for N binary variables
- If each variable has M different choices, $M^N - 1$ parameters are required

- The number of parameters grow exponentially

→ **Difficult to represent probability distribution and learn the parameters from data**

Motivation of Bayesian network

- Full joint Probability Distribution



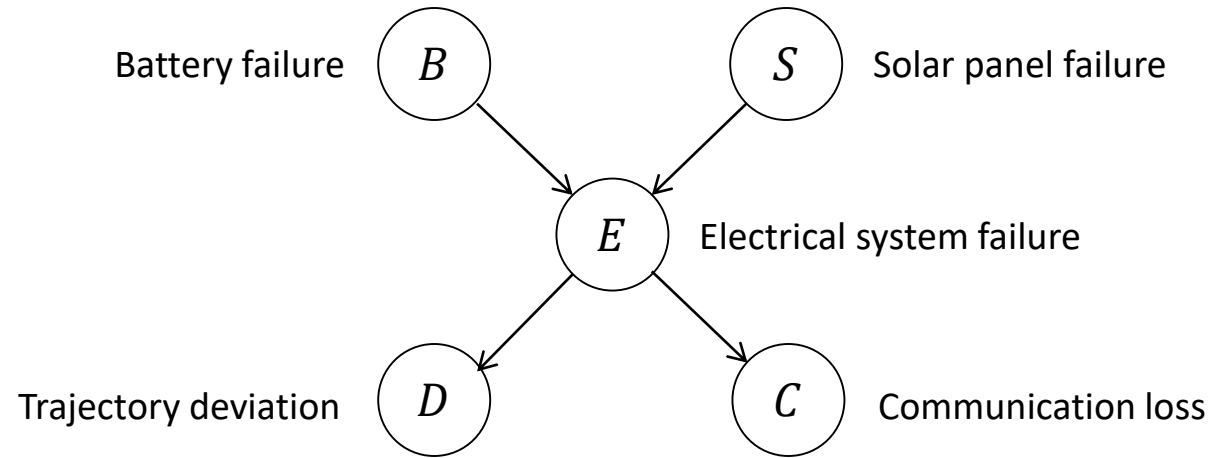
- Binary variables: B, S, E, D, C (e.g., $B = 1$ or 0)
- 2^5 entities are required to construct the table
- $2^5 - 1$ independent parameters are required to fully specify the joint probability distribution
- $2^N - 1$ parameters are required for N binary variables
- If each variables has M different choices, $M^N - (M - 1)$ parameters are required

The number of parameters grows exponentially

→ **Difficult to represent Probability distribution and learn the parameters from data**

Motivation of Bayesian network

- A Bayesian network is a compact representation of a joint distribution

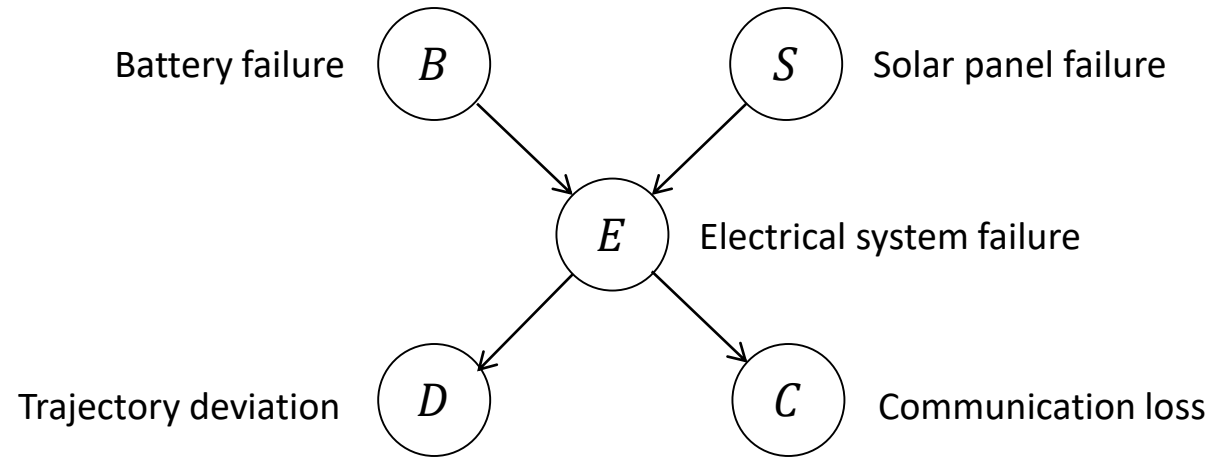


Probability + **Statistics** + **Graph Theory**

- A **Bayesian network** introduces structure into a probabilistic model by using graphs to represent independence assumptions among the variables.
- Provide a good representation to model the probabilistic structures between random variables.
 - Node represent random variables
 - Edge represent probabilistic dependency, namely conditional probability among variables
- Conditional independence described by the graph, greatly reduce the computational effort to learn the model and inferencing random variables.

Motivation of Bayesian network

- A Bayesian network is a compact representation of a joint distribution



- Each node corresponds to a random variable
- Directed edges connect pairs of nodes, indicating direct probabilistic relationships
- $P(x_i | \text{pa}_{x_i})$ represent the probability distribution of x_i conditional on the parent nodes pa_{x_i} of x_i e.g., $P(E|B, S)$: *B* and *S* are the parent nodes of *E*

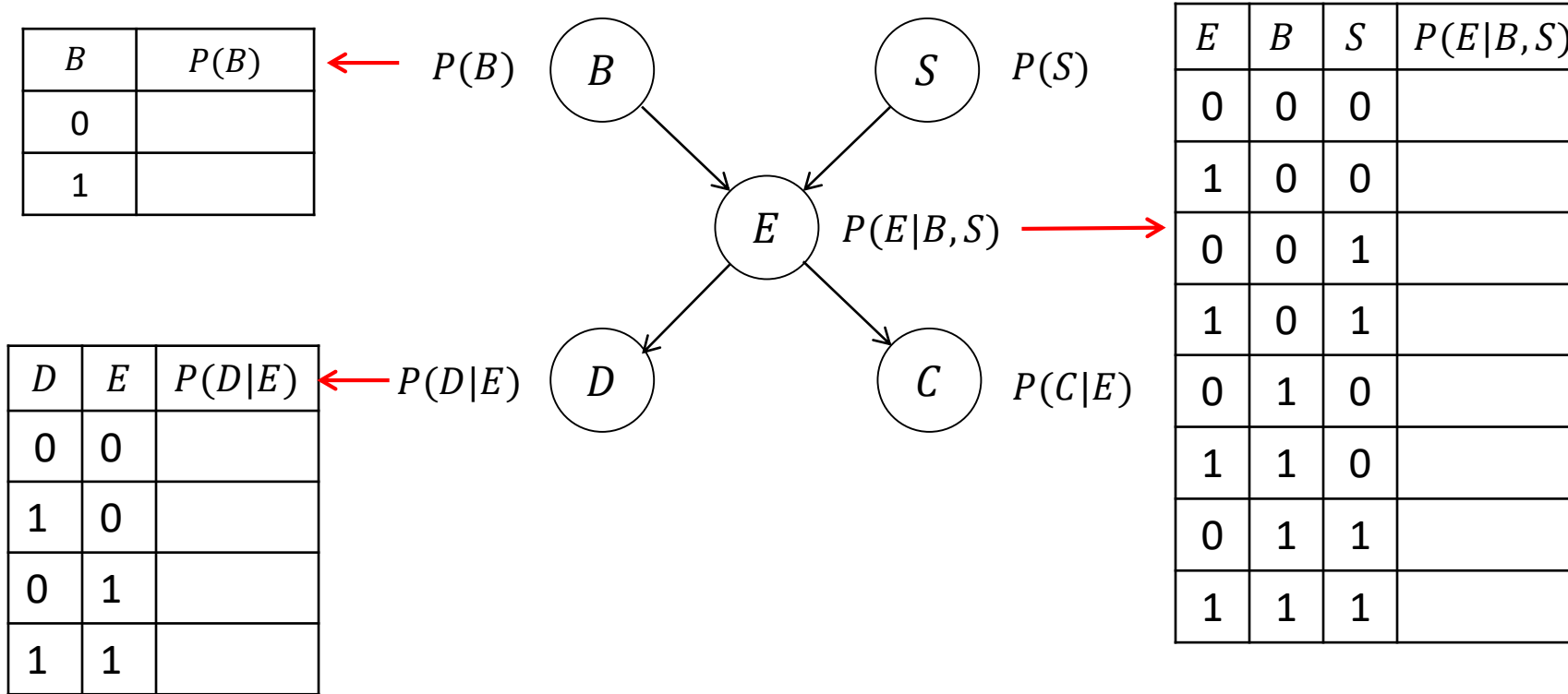
The chain rule for Bayesian networks specifies how to construct a joint distribution from the local conditional probability distribution

$$P(x_1, \dots, x_n) = \prod_{i=1}^n P(x_i | \text{pa}_{x_i})$$

local conditional probability distribution

Motivation of Bayesian network

- A Bayesian network is a compact representation of a joint distribution



- Chain rule: $P(B, S, E, D, C) = P(B)P(S)P(E|B, S)P(D|E)P(C|E)$
- Required independent parameters to fully specify the joint PDF

$P(B): 1, P(S): 1, P(E|B, S): 4, P(D|E): 2, P(C|E): 2$ (total 10 compared to $2^5 - 1 = 31$)

Bayesian network can greatly reduce the number of parameters

Formal definition of Bayesian network

- A Bayesian network (BN) is a distribution of the form

$$p(x_1, \dots, x_n) = \prod_{i=1}^n p(x_i | \text{pa}_{x_i})$$

- ✓ pa_{x_i} represents the parental variables of variable x_i
 - ✓ BN is represented as a directed acyclic graph with an arrow pointing from a parent variable to child variable
- Every probability distribution can be written as a BN:

$$\begin{aligned} p(x_1, \dots, x_n) &= p(x_n | x_1, \dots, x_{n-1}) p(x_1, \dots, x_{n-1}) \\ &= p(x_n | x_1, \dots, x_{n-1}) p(x_{n-1} | x_1, \dots, x_{n-2}) p(x_1, \dots, x_{n-2}) \\ &= p(x_1) \prod_{i=2}^n p(x_i | \text{pa}_{x_{i-1}}) \end{aligned}$$

- The particular role of BN is that the structure of the DAG corresponds to a set of **conditional independence assumptions**, namely which ancestral parental variables are sufficient to specify each conditional probability table

Conditional independence

- Definition of **independence**

$$X \perp Y$$

$$p(X, Y) = p(X)p(Y) \text{ for all states of } X, Y$$

or equivalently $p(X|Y) = p(X)$

- Definition of **conditional independence**

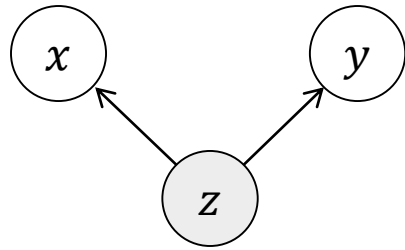
$$X \perp Y|Z$$

$$p(X, Y|Z) = p(X|Z)p(Y|Z) \text{ for all states of } X, Y, Z$$

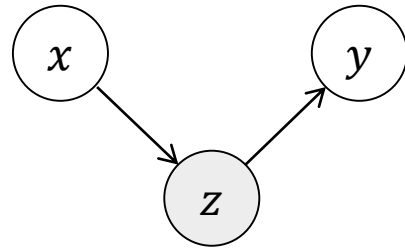
or equivalently $p(X|Y, Z) = p(X|Z)$

- ✓ The two sets of variables X and Y are independent of each other provided we know the state of the set of variables Z
- ✓ The information of Y does not give further information on X

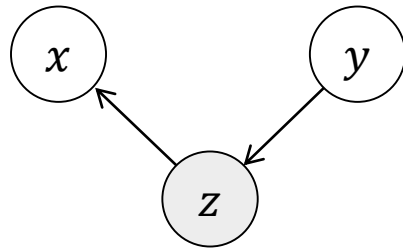
V-structure (or collider)



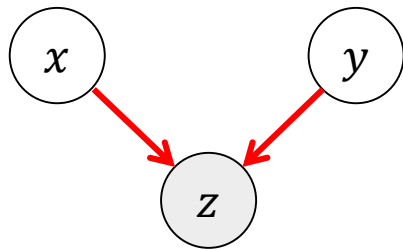
$$p(x, y|z) = \frac{p(x, y, z)}{p(z)} = \frac{p(z)p(x|z)p(y|z)}{p(z)} = p(x|z)p(y|z)$$



$$\begin{aligned} p(x, y|z) &= \frac{p(x, y, z)}{p(z)} = \frac{p(x)p(z|x)p(y|z)}{p(z)} \\ &= \frac{p(x, z)p(y|z)}{p(z)} = p(x|z)p(y|z) \end{aligned}$$



$$\begin{aligned} p(x, y|z) &= \frac{p(x, y, z)}{p(z)} = \frac{p(y)p(z|y)p(x|z)}{p(z)} \\ &= \frac{p(y, z)p(x|z)}{p(z)} = p(y|z)p(x|z) \end{aligned}$$

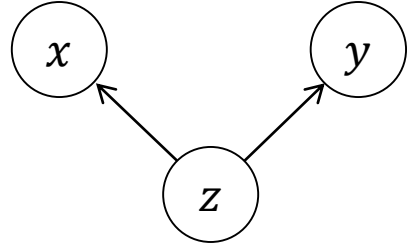


$$p(x, y|z) = \frac{p(x, y, z)}{p(z)} = \frac{p(x)p(y)p(z|x, y)}{p(z)} \neq p(y|z)p(x|z)$$

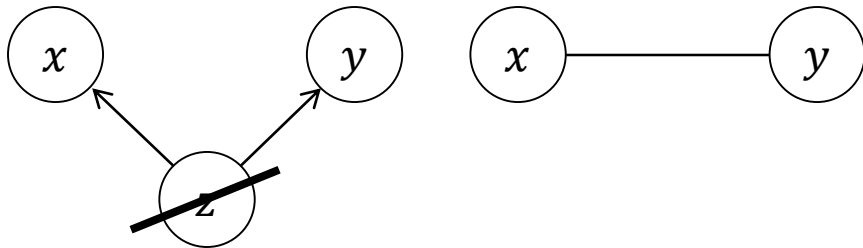
- BN with $x \rightarrow z \leftarrow y$
- ✓ x and y are unconditionally independent
 - ✓ x and y are dependent conditional on z

V-structure (or collider)

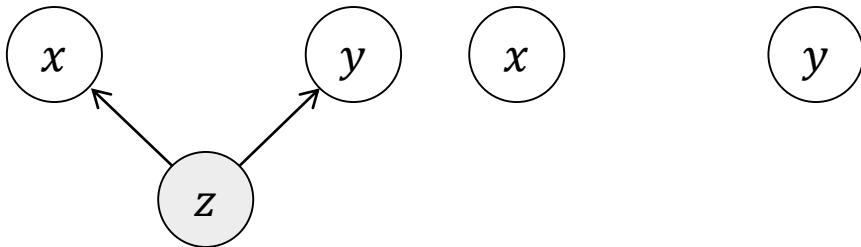
$$p(x, y, z) = p(x|z)p(y|z)$$



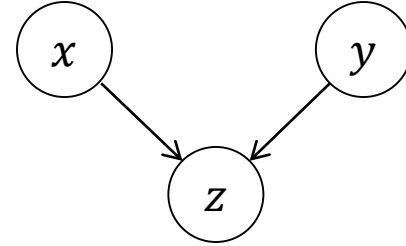
Marginalization over z



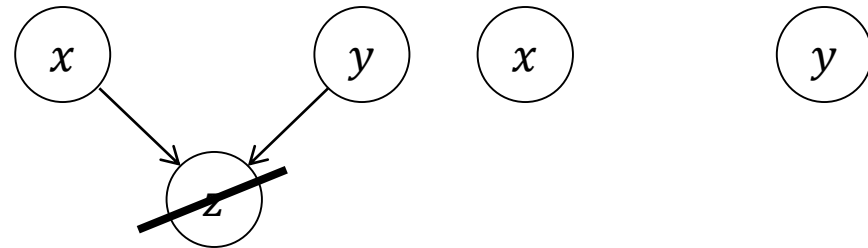
Conditionalization on z



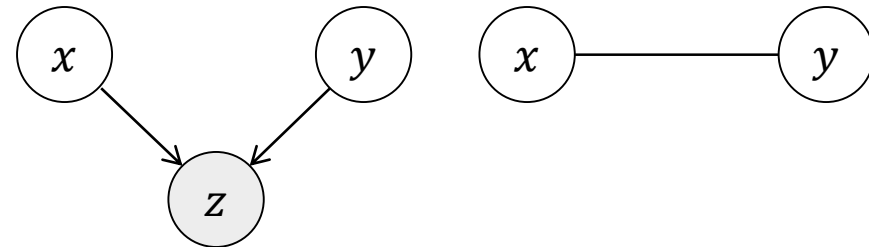
$$p(x, y, z) = p(z|x, y)p(x)p(y)$$



Marginalization over z

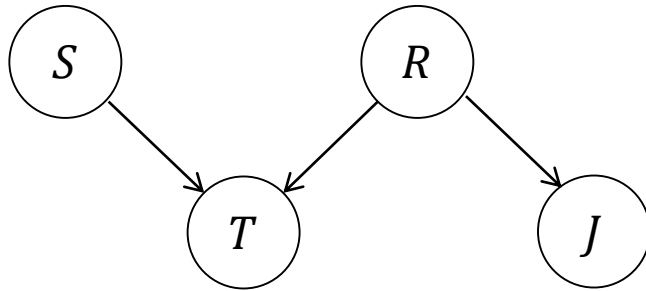


Conditionalization on z



Conditional independence

Example 3.1 (Wet Grass)



$R \in \{0,1\} : R = 1$ means that it has been raining

$S \in \{0,1\} : S = 1$ Sprinkler is turned on

$J \in \{0,1\} : J = 1$ Jack's grass is wet

$T \in \{0,1\} : T = 1$ Tracey's grass is wet

Joint distribution based on chain rule

$$p(T, J, R, S) = p(T|J, R, S)p(J, R, S)$$

$$= p(T|J, R, S)p(J|R, S)p(R, S)$$

$$= p(T|J, R, S)p(J|R, S)p(R|S)p(S)$$

$$8 + 4 + 2 + 1 = 2^4 - 1 = 15 \text{ parameters are required}$$

Joint distribution conditional independence

$$p(T, J, R, S) = p(T|J, R, S)p(J|R, S)p(R|S)p(S)$$

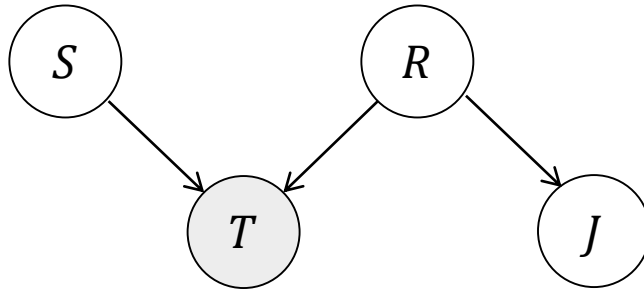
$$= p(T|R, S) \times p(J|R) \times p(R) \times p(S)$$

$$= p(T|R, S)p(J|R)p(R)p(S)$$

Conditional independence

Example 3.1 (Wet Grass)

Modeling



$R \in \{0,1\}$: $R = 1$ means that it has been raining

$S \in \{0,1\}$: $S = 1$ Sprinkler is turned on

$J \in \{0,1\}$: $J = 1$ Jack's grass is wet

$T \in \{0,1\}$: $T = 1$ Tracey's grass is wet

$$p(T, J, R, S) = p(T|R, S)p(J|R)p(R)p(S)$$

$p(T|S, R)$

Tracey's Grass wet=1	Rain	Sprinkler
1	1	1
1	1	0
0.9	0	1
0	0	0

$p(J|R)$

Jack's Grass wet=1	Rain
1	1
0.2	0

$$p(S = 1) = 0.1$$

$$p(R = 1) = 0.2$$

The tables and graphical structure fully specify the distribution

Conditional independence

Example 3.1 (Wet Grass)

Inference

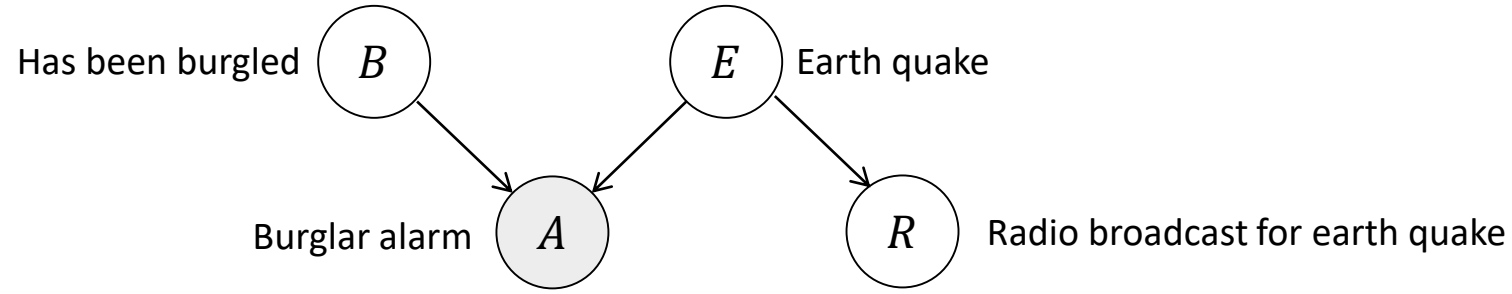
$$\begin{aligned} p(S = 1|T = 1) &= \frac{p(S = 1, T = 1)}{p(T = 1)} = \frac{\sum_{J,R} p(T = 1, J, R, S = 1)}{\sum_{J,R,S} p(T = 1, J, R, S)} \\ &= \frac{\sum_{J,R} p(J|R) p(T = 1|R, S = 1) p(R) p(S = 1)}{\sum_{J,R,S} p(J|R) p(T = 1|R, S) p(R) p(S)} \\ &= \frac{\sum_R p(T = 1|R, S = 1) p(R) p(S = 1)}{\sum_{R,S} p(T = 1|R, S) p(R) p(S)} \quad \because \sum_J p(J|R) = 1 \\ &= \frac{0.9 \times 0.8 \times 0.1 + 1 \times 0.2 \times 0.1}{0.9 \times 0.8 \times 0.1 + 1 \times 0.2 \times 0.1 + 0 \times 0.8 \times 0.9 + 1 \times 0.2 \times 0.9} = 0.3382 \end{aligned}$$

$$\begin{aligned} p(S = 1|T = 1, J = 1) &= \frac{p(S = 1, T = 1, J = 1)}{p(T = 1, J = 1)} \\ &= \frac{\sum_R p(T = 1, J = 1, R, S = 1)}{\sum_{R,S} p(T = 1, J = 1, R, S)} \\ &= \frac{\sum_R p(J = 1|R) p(T = 1|R, S = 1) p(R) p(S = 1)}{\sum_{R,S} p(J = 1|R) p(T = 1|R, S) p(R) p(S)} \\ &= \frac{0.0344}{0.2144} = 0.1604 \end{aligned}$$

The fact that Jack's grass is also wet increases the chance that the rain has played a role in making Tracey's grass wet

Conditional independence

Example 3.2 (Burglar Alarm)



$$p(B, E, A, R) = p(A|B, E)p(R|E)p(E)p(B)$$

$$p(A|B, E)$$

Alarm=1	Burglar	Earthquake
0.9999	1	1
0.99	1	0
0.99	0	1
0.0001	0	0

$$p(R|E)$$

Radio=1	Earthquake
1	1
0	0

$$p(E = 1) = 0.01$$

$$p(E = 0) = 0.99$$

$$\begin{aligned} p(B = 1|A = 1) &= \frac{p(B, A = 1)}{p(A = 1)} = \frac{\sum_{E,R} p(B = 1, E, A = 1, R)}{\sum_{B,E,R} p(B, E, A = 1, R)} \\ &= \frac{\sum_{E,R} p(A = 1|B = 1, E)p(R|E)p(E)p(B = 1)}{\sum_{B,E,R} p(A = 1|B, E)p(R|E)p(E)p(B)} \approx 0.99 \\ p(B = 1|A = 1, R = 1) &\approx 0.01 \end{aligned}$$

Conditional independence

- Where causes the number of parameters to be reduced?
→ **The conditional independence assumptions** encoded by the structure of a Bayesian network
- X and Y are independent if and only if

$$P(X, Y) = P(X)P(Y)$$

$$\text{or equivalently } P(X|Y) = P(X)$$

$$\because P(X|Y) = \frac{P(X, Y)}{P(Y)} = \frac{P(X)P(Y)}{P(Y)} = P(X)$$

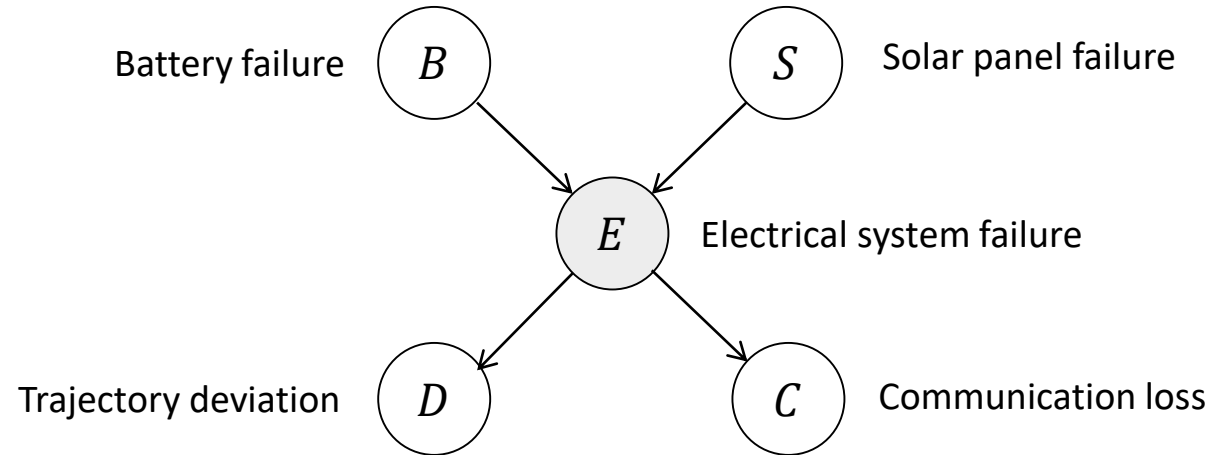
- X and Y are conditionally independent given Z if and only if

$$P(X, Y|Z) = P(X|Z)P(Y|Z)$$

$$\text{or equivalently } P(X|Z) = P(X|Y, Z)$$

Independence assumptions reduce the number of parameters used to represent a joint pdf

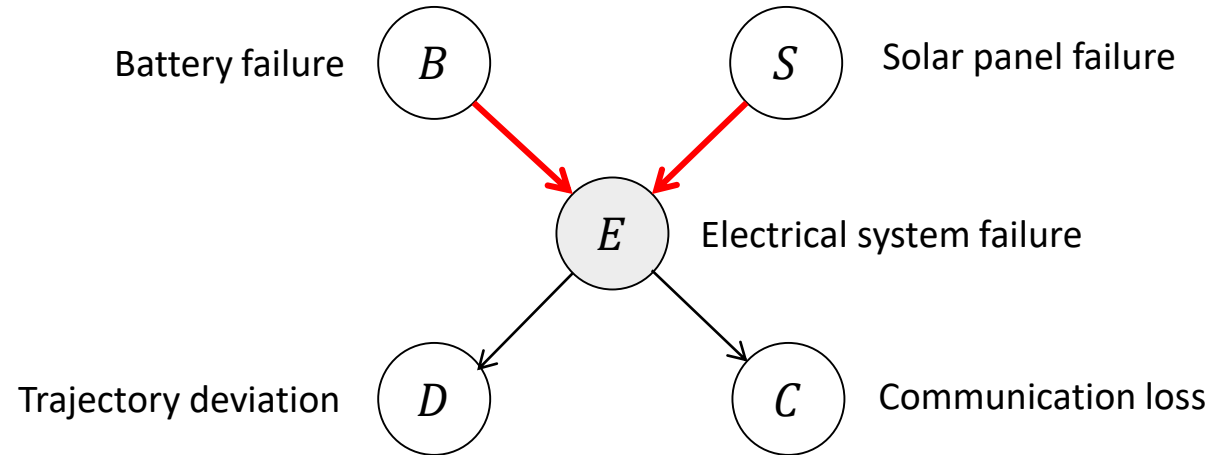
Conditional independence



- C is independent of B given E : $(C \perp B|E)$
→ Information about Battery failure does not affect my belief on communication loss if I already know (observed) the status of electrical system failure
- D is independent of S given E : $(D \perp S|E)$
→ Information about Solar failure does not affect my belief on a trajectory deviation if I already know (observed) the status of electrical system failure

Conditional independence

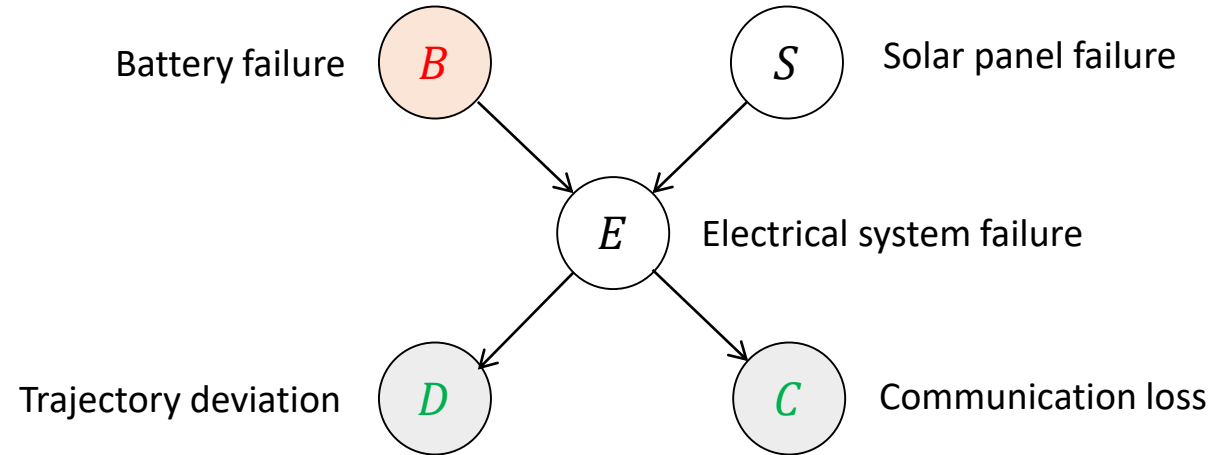
V-structure



- B is independent S (E is not observed)
→ Knowing there is a battery failure does not affect my belief regarding solar panel failure
- B is independent S **given E**
→ If there was an electrical system failure (observed) and there was no battery failure, there it is likely that a solar panel fails
- Influence flows only through $B \rightarrow E \leftarrow S$ when E is known

Inference

- Once a joint probability is constructed, inference can be performed to determine the distribution over one or more unobserved variables given the values associated with a set of observed variables



- $P(B|d^1, c^1)$: Probability distribution of Battery failure
Query variable
given the trajectory deviation and the communication loss
Evidence variable

E *S* : Hidden variables

Inference

- How to compute $P(B|d^1, c^1)$?
- **Exact inference**

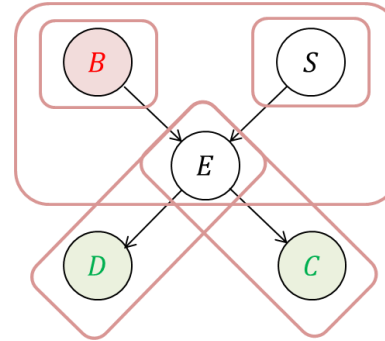
$$\begin{aligned} P(b^1|d^1, c^1) &\propto \sum_s \sum_e P(b^1, s, e, d^1, c^1) \\ &= \sum_s \sum_e P(b^1)P(s)P(e|b^1, s)P(d^1|e)p(c^1|e) && \text{By conditional independence} \\ &= P(b^1) \sum_e P(d^1|e)p(c^1|e) \sum_s P(s)P(e|b^1, s) \end{aligned}$$

- The number of terms to be added together can grow exponentially with the number of hidden variables

Inference

- How to compute $P(B|d^1, c^1)$?

- Variable Elimination**



- Conditional distributions are represented by the following tables

$$T_1(B)T_2(S)T_3(E, B, S)T_4(d^1, E)T_5(c^1, E)$$

$$T_1(B)T_2(S)T_3(E, B, S)T_6(E)T_7(E)$$

Observe evidence (d^1 and c^1)

$$T_1(B)T_2(S)T_8(B, S)$$

$$T_8(B, S) = \sum_e T_3(e, B, S)T_6(e)T_7(e)$$

$$T_1(B)T_9(B)$$

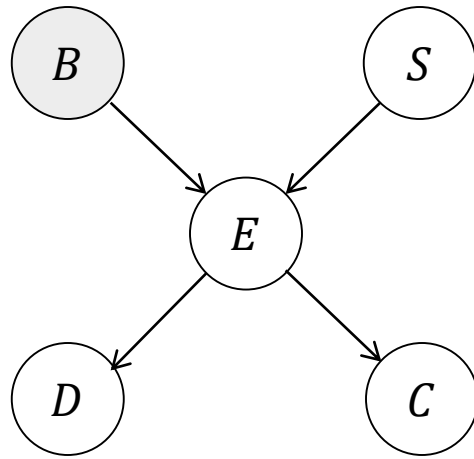
$$T_9(B) = \sum_s T_2(s)T_8(B, s)$$

- Normalizing the product of the two factors ($T_1(B)$ and $T_9(B)$) results in $P(B|d^1, c^1)$

Variable elimination algorithm relies on **heuristic ordering** of variables to eliminate in sequence
 → Often linear but sometimes exponential

Inference

- How to compute $P(B|d^1, c^1)$?
- **Approximate inference (Sampling based methods)**

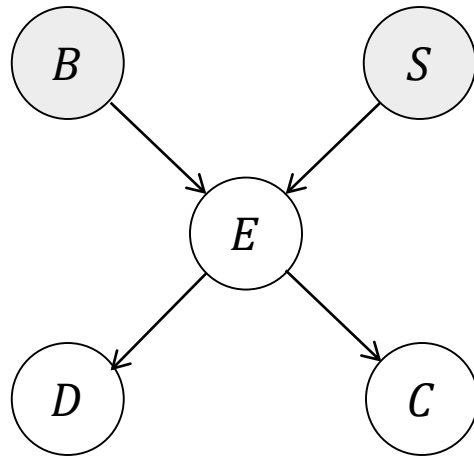


Sample from $P(B)$

B	S	E	D	C
1				

Inference

- How to compute $P(B|d^1, c^1)$?
- **Approximate inference (Sampling based methods)**

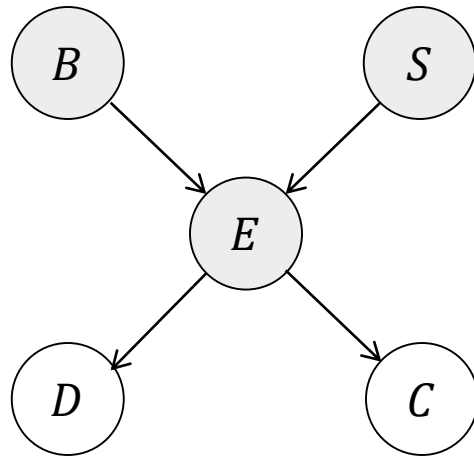


Sample from $P(S)$

B	S	E	D	C
1	1			

Inference

- How to compute $P(B|d^1, c^1)$?
- **Approximate inference (Sampling based methods)**

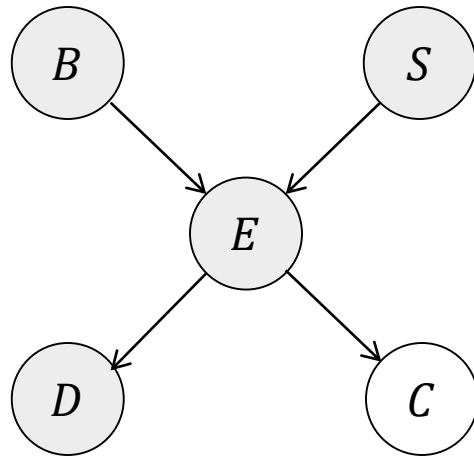


B	S	E	D	C
1	1	1		

Sample from $P(E|B = 1, S = 1)$

Inference

- How to compute $P(B|d^1, c^1)$?
- **Approximate inference (Sampling based methods)**

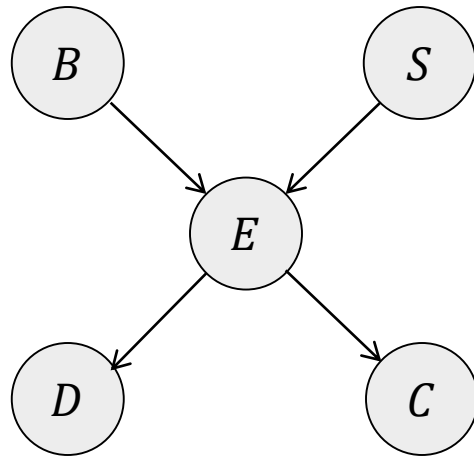


Sample from $P(D|E = 1)$

B	S	E	D	C
1	1	1	0	

Inference

- How to compute $P(B|d^1, c^1)$?
- **Approximate inference (Sampling based methods)**

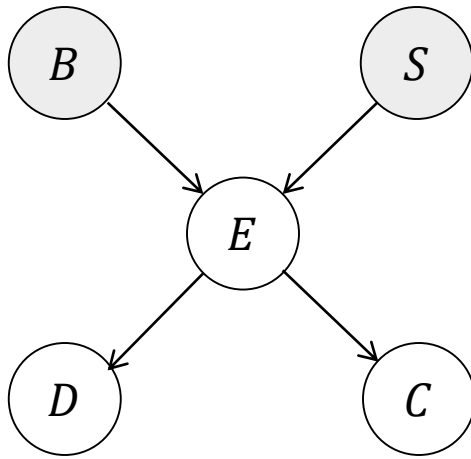


B	S	E	D	C
1	1	1	0	0

Sample from $P(C|E = 1)$

Inference

- How to compute $P(B|d^1, c^1)$?
- Approximate inference (Sampling based methods)**



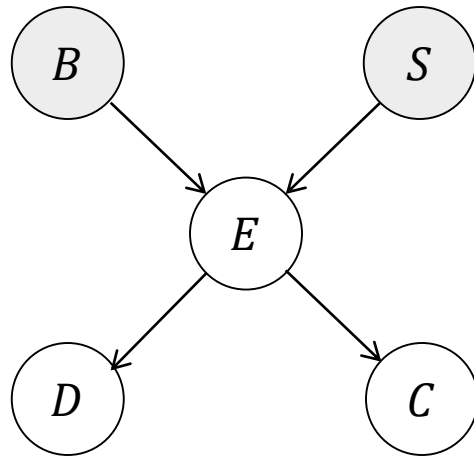
$$P(b^1|d^1, c^1) = 1/3$$
$$P(b^0|d^1, c^1) = 2/3$$

B	S	E	D	C
1	1	1	0	0
0	1	0	1	0
1	0	1	1	1
0	1	0	0	1
0	1	1	1	1
0	1	0	0	1
0	0	0	1	0
0	1	1	1	0
0	1	0	1	1

Three cases coincide observations d^1, c^1

Inference

- How to compute $P(B|d^1, c^1)$?
- Approximate inference (Sampling based methods)**



$$P(b^1|d^1, c^1) = 1/3$$
$$P(b^0|d^1, c^1) = 2/3$$

B	S	E	D	C
1	1	1	0	0
0	1	0	1	0
1	0	1	1	1
0	1	0	0	1
0	1	1	1	1
0	1	0	0	1
0	0	0	1	0
0	1	1	1	0
0	1	0	1	1

Three cases coincide observations d^1, c^1

If likelihood of evidence is small, then many samples are required!!

Inference

- How to compute $P(B|d^1, c^1)$?
- Likelihood sampling**

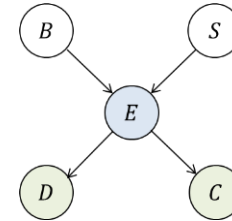
B	S	E	D	C
1	1	1	0	0
0	1	0	1	0
1	0	1	1	1
0	1	0	0	1
0	1	1	1	1
0	1	0	0	1
0	0	0	1	0
0	1	1	1	0
0	1	0	1	1

Algorithm 2.5 Likelihood-weighted sampling from a Bayesian network

```

1: function LIKELIHOODWEIGHTEDSAMPLE( $B, o_{1:n}$ )
2:    $X_{1:n} \leftarrow$  a topological sort of nodes in  $B$ 
3:    $w \leftarrow 1$ 
4:   for  $i \leftarrow 1$  to  $n$ 
5:     if  $o_i = \text{NIL}$ 
6:        $x_i \leftarrow$  a random sample from  $P(X_i | \text{pa}_{x_i})$ 
7:     else
8:        $x_i \leftarrow o_i$ 
9:        $w \leftarrow w \times P(x_i | \text{pa}_{x_i})$ 
10:  return  $(x_{1:n}, w)$ 

```



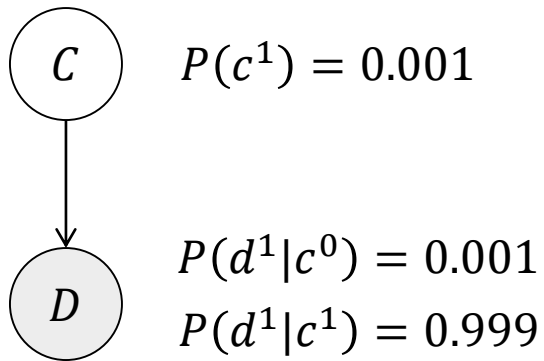
B	S	E	D	C	weight
1	0	1	1	1	$P(d^1 e^1) P(c^1 e^1)$
0	1	1	1	1	$P(d^1 e^1) P(c^1 e^1)$
0	1	0	1	1	$P(d^1 e^0) P(c^1 e^0)$

$$P(b^1|d^1, c^1) = \frac{P(d^1|e^1) P(c^1|e^1)}{P(d^1|e^1) P(c^1|e^1) + P(d^1|e^1) P(c^1|e^1) + P(d^1|e^0) P(c^1|e^0)}$$

Inference

- How to compute $P(B|d^1, c^1)$?

Likelihood sampling has a still problem!



Bayesian approach :

$$\begin{aligned} P(c^1|d^1) &= \frac{P(d^1|c^1)P(c^1)}{P(d^1|c^1)P(c^1) + P(d^1|c^0)P(c^0)} \\ &= \frac{0.999 \times 0.001}{0.999 \times 0.001 + 0.001 \times 0.999} \\ &= 0.5 \end{aligned}$$

To use likelihood weighting sampling approach:

$c^0, c^0, c^0, c^0, c^0, c^0, c^0, c^0, c^0, c^0, c^0, c^0, c^0, c^0, \dots, c^1$

$P(d^1|c^1) = 0$ because c^1 is not sampled due to the low prior

Inference

- How to compute $P(B|d^1, c^1)$?
- **Gibbs sampling, a kind of Markov chain Monte Carlo technique**
- The sequence of samples from a Markov chain
- In the limit, samples are drawn exactly from the joint distribution over the unobserved variables given the observations
- Simulate sample by sweeping through all the posterior conditionals, one random variables at a time

Algorithm : Gibbs sampler

Initialize $X^{(0)} \sim q(x)$

for iteration $i = 1, \dots$ *do*

$$x_1^{(i)} \sim P\left(X_1 = x_1 \mid X_2 = x_2^{(i-1)}, X_3 = x_3^{(i-1)}, \dots, X_D = x_D^{(i-1)}\right)$$

$$x_2^{(i)} \sim P\left(X_2 = x_2 \mid X_1 = x_1^{(i)}, X_3 = x_3^{(i-1)}, \dots, x_D = x_D^{(i-1)}\right)$$

$$x_3^{(i)} \sim P\left(X_3 = x_3 \mid X_1 = x_1^{(i)}, X_2 = x_2^{(i)}, \dots, x_D = x_D^{(i-1)}\right)$$

$$\vdots$$

$$x_D^{(i)} \sim P\left(X_D = x_D \mid X_1 = x_1^{(i)}, X_2 = x_2^{(i)}, \dots, X_{D-1} = x_{D-1}^{(i)}\right)$$

end for

- Because the samples from the early iterations are not from the target posterior, it is common to discard these samples **“burn-in” period**

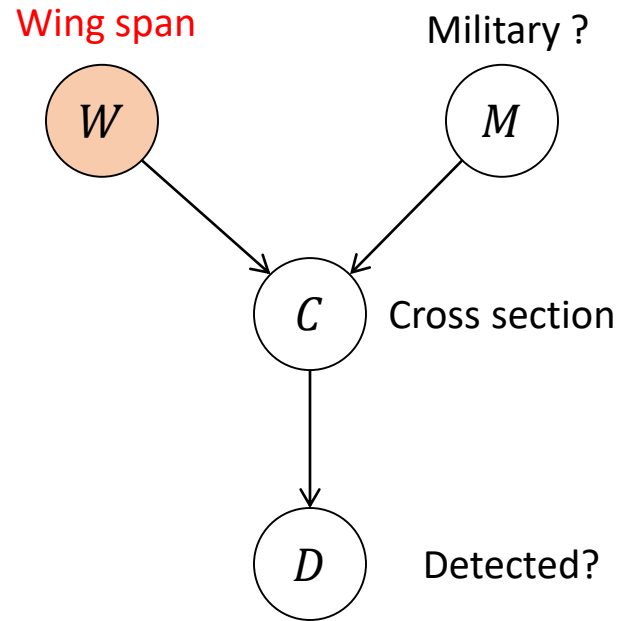
Inference

- **Sampling method comparisons**

Jupyter Demo Simulation
Wet grass (PyMC)

Hybrid Bayesian networks

- Bayesian networks can contain a mixture of both discrete and continuous variables

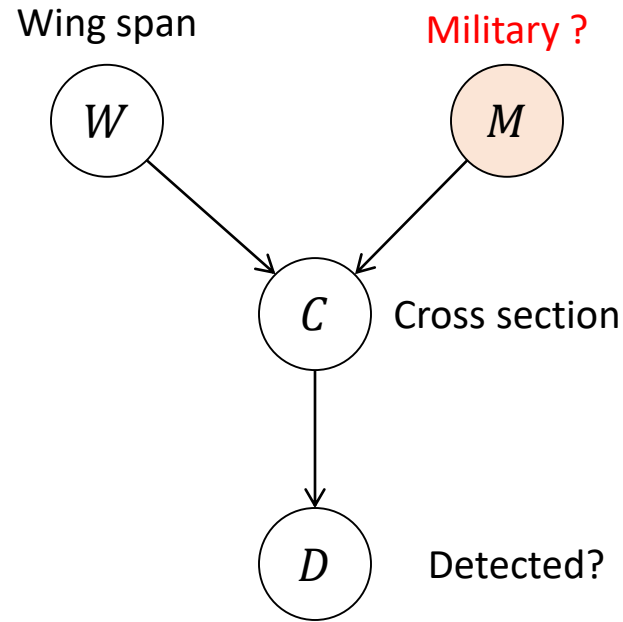


Wing span is a continuous variable and modeled as a Gaussian distribution

$$P(w) = N(w|\mu, \sigma^2) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{w-\mu}{\sigma}\right)^2}$$

Hybrid Bayesian networks

- Bayesian networks can contain a mixture of both discrete and continuous variables



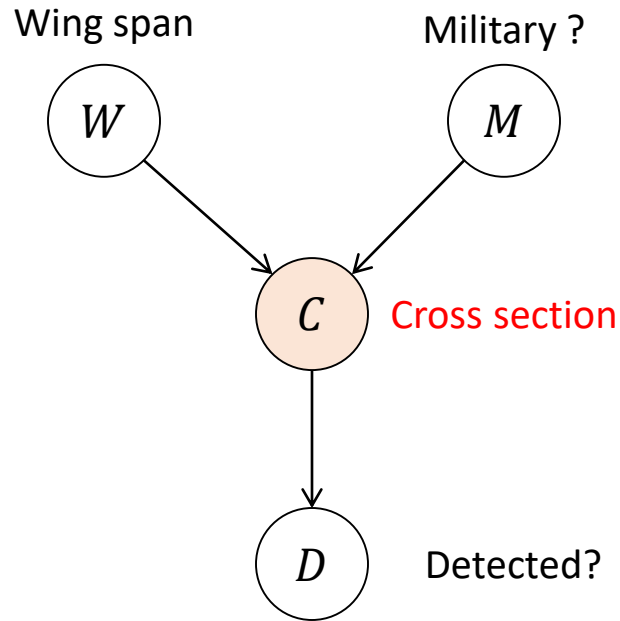
Whether a target is a military vehicle can be modeled with a single parameter θ

$$P(m^1) = \theta$$

$$P(m^0) = 1 - \theta$$

Hybrid Bayesian networks

- Bayesian networks can contain a mixture of both discrete and continuous variables



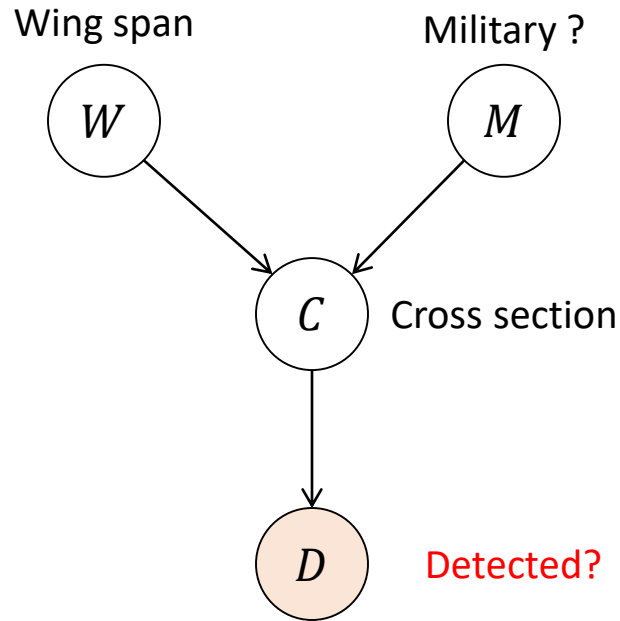
Radar cross section can be modeled as a conditional Gaussian

$$P(c|w, m) = \begin{cases} N(c|a_0 w + b_0, \sigma_0^2) & \text{if } m = m^0 \\ N(c|a_1 w + b_1, \sigma_1^2) & \text{if } m = m^1 \end{cases}$$

(Conditional linear Gaussian)

Hybrid Bayesian networks

- Bayesian networks can contain a mixture of both discrete and continuous variables

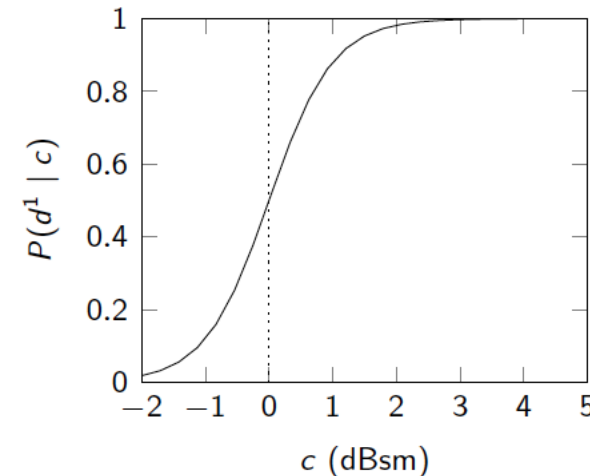


- Logit model:

$$P(d^1|c) = \frac{1}{1 + \exp\left(-2\frac{c - \alpha}{\beta}\right)}$$

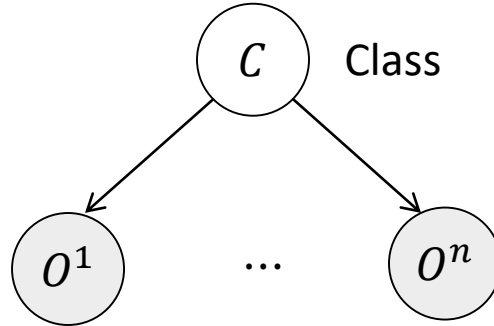
- Probit model:

$$P(d^1|c) = \Phi\left(\frac{c - \alpha}{\beta}\right)$$



Bayesian network for Classification

- Naïve Bayes Model



- Prior: $P(C)$
- Class conditional distribution: $P(O^i|C)$

$$P(C|O^{1:n}) = \frac{P(C, O^{1:n})}{P(O^{1:n})} = \frac{P(C) \prod_{i=1}^n P(O^i|C)}{P(O^{1:n})}$$

$$P(O^{1:n}) = \sum_c P(C, O^{1:n})$$

$$P(C|O^{1:n}) \propto P(C) \prod_{i=1}^n P(O^i|C)$$

Parameter learning

We already know how to estimate the parameters for probability distributions

MLE or Bayesian approach

Structure learning

- Bayesian Score $P(G|D)$ for a certain graph G given data D is defined as

$$\begin{aligned} P(G|D) &= \frac{P(G)P(D|G)}{P(D)} \\ &= \frac{P(G) \int_{\theta} P(D|\theta, G)P(\theta|G)d\theta}{P(D)} \end{aligned}$$

- A Bayesian approach to structure learning involves finding the graph G that maximizes the Bayesian Score $P(G|D)$ as

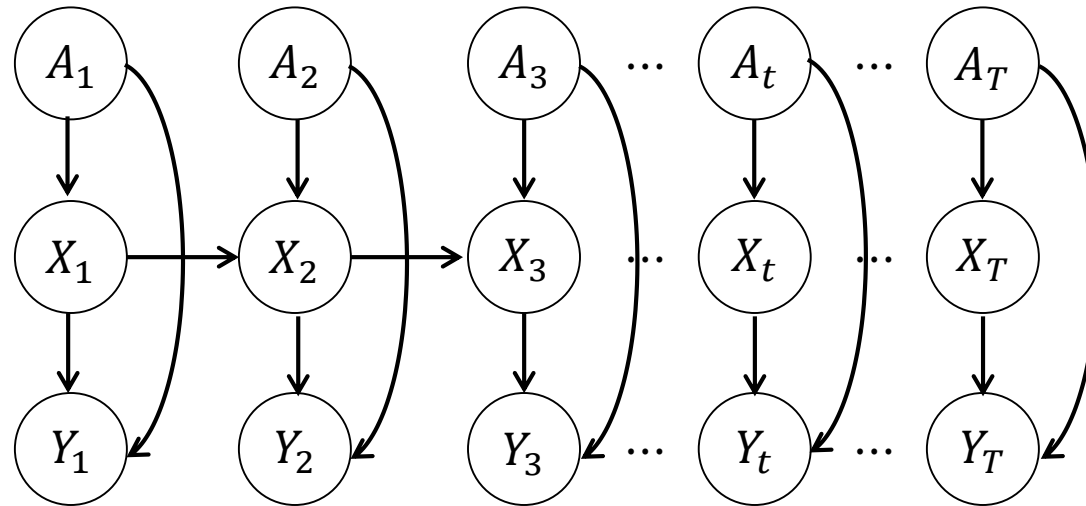
$$G^* = \operatorname{argmax}_G P(G|D)$$

- Not feasible to enumerate every possible structure, so use local search for graph with largest Bayesian score

Dynamic Bayesian Network

Motivation

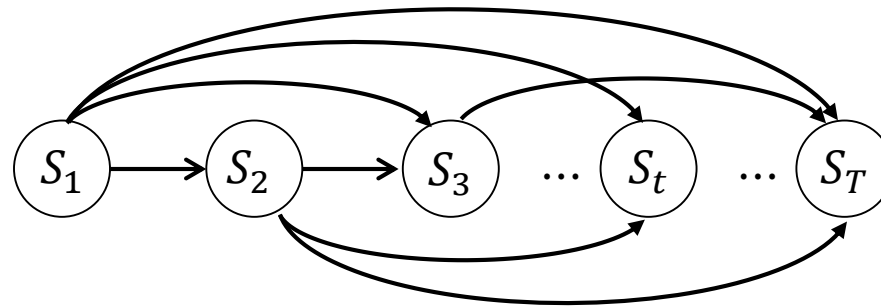
Dynamic Bayesian Network relates variables to each other **over adjacent time steps**.



Dynamic Bayesian networks (temporal model)

- We are interested in reasoning about the state of the world as it evolves over time
- **System state** S_t is a snapshot of the relevant attributes of the system at time t
- Trajectory of states $S_1, \dots, S_t$ represents the evolution of the target system
- $P(S_1, \dots, S_t)$ is very complex probability space

→ we need a series of simplifying assumptions

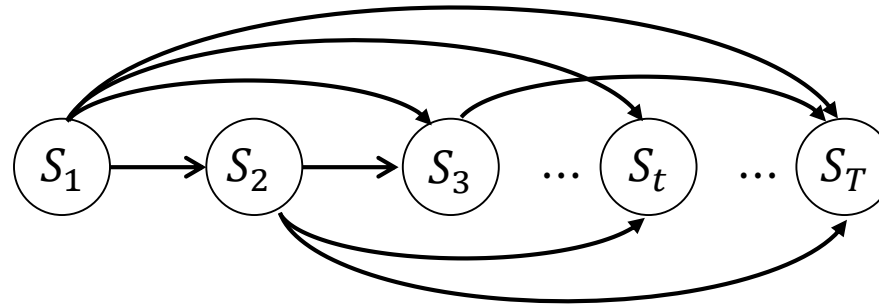


Discrete-state Markov models

- Consider a distribution over trajectories sampled over a prefix of time $t = 1, \dots, T$

$$P(S_1, S_2, \dots, S_T) = P(S_1) \prod_{t=2}^T P(S_t | S_{1:t-1})$$

Cascade decomposition



Discrete-state Markov models

- Consider a distribution over trajectories sampled over a prefix of time $t = 1, \dots, T$

$$P(S_1, S_2, \dots, S_T) = P(S_1) \prod_{t=2}^T P(S_t | S_{1:t-1}) \quad \text{Cascade decomposition}$$

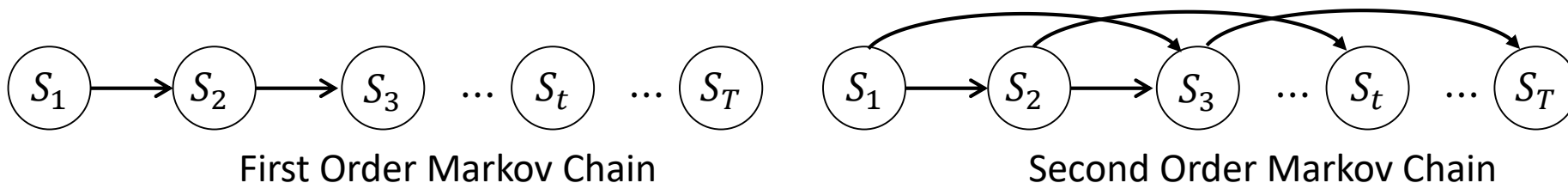
1. Markov Chain: A Markov chain is defined on either discrete or continuous variables $S_{1:t}$ and the following **conditional independence** assumption holds:

$$P(S_t | S_1, \dots, S_{t-1}) = P(S_t | S_{t-L}, \dots, S_{t-1})$$

where $L \geq 1$ is the order of the Markov chain. First order Markov chain can be represented

$$P(S_1, S_2, \dots, S_T) = P(S_1) \prod_{t=2}^T P(S_t | S_{t-1})$$

the future is conditionally independent of the pass given the present: $(S_{t+1} \perp S_{(1:t-1)} | S_t)$



Discrete-state Markov models

- Consider a distribution over trajectories sampled over a prefix of time $t = 1, \dots, T$

$$P(S_1, S_2, \dots, S_T) = P(S_1) \prod_{t=2}^T P(S_t | S_{1:t-1}) \quad \text{Cascade decomposition}$$

1. Markov Chain: A Markov chain is defined on either discrete or continuous variables $S_{1:t}$ and the following **conditional independence** assumption holds:

$$P(S_t | S_1, \dots, S_{t-1}) = P(S_t | S_{t-L}, \dots, S_{t-1})$$

where $L \geq 1$ is the order of the Markov chain. First order Markov chain can be represented

$$P(S_1, S_2, \dots, S_T) = P(S_1) \prod_{t=2}^T P(S_t | S_{t-1})$$

2. Stationary assumption: The state transition probability $P(S_{t+1} | S_t)$ is the same for all t

$$P(S_{t+1} = s' | S_t = s) = P(S' = s' | S = s), \quad \forall t$$

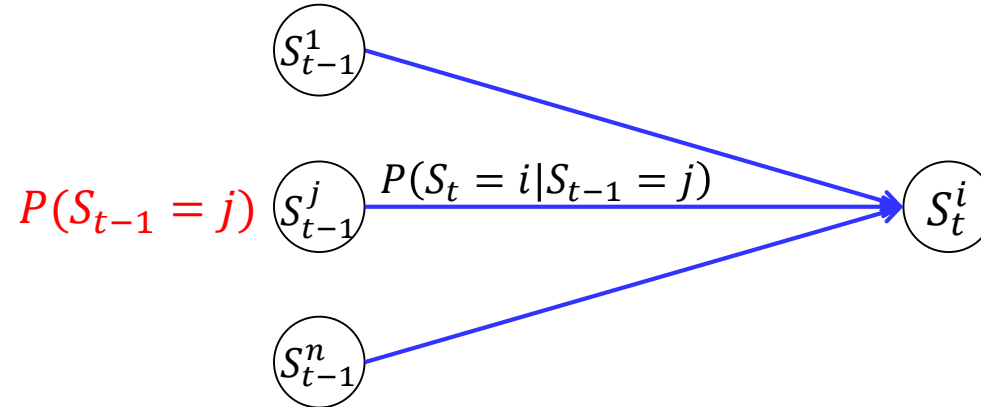
→ the number of parameters are reduced substantially

Equilibrium and stationary distribution of a Markov chain

- The marginal $P(S_t)$ evolves through time. For discrete time,

$$P(S_t = i) = \sum_j P(S_t = i | S_{t-1} = j) P(S_{t-1} = j)$$

- ✓ $P(S_t = i)$: the frequency that we visit state i at time t , given we started with a sample from $P(S_1)$ and subsequently repeatedly drew samples from the transition $P(S_\tau | S_{\tau-1})$



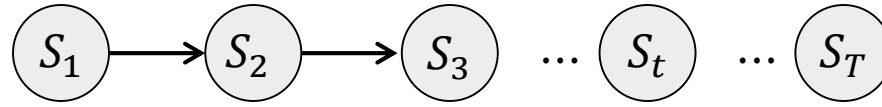
- Denoting $(\mathbf{p}_t)_i = P(S_t = i)$,

$$\mathbf{p}_t = \mathbf{M}\mathbf{p}_{t-1} = \mathbf{M}^{t-1}\mathbf{p}_1$$

- If for $t \rightarrow \infty$, $\mathbf{p}_t$ is independent of the initial distribution of $\mathbf{p}_1$, then $\mathbf{p}_\infty$ is called the equilibrium distribution of the chain, that is

$$\mathbf{p}_\infty = \mathbf{M}\mathbf{p}_\infty$$

Fitting Markov Models



Given a sequence $(S_1 = s_1, S_2 = s_2, \dots, S_T = s_T)$, how to construct the transition matrix?

$$\theta_{i|j} = P(S_\tau = i | S_{\tau-1} = j) \propto \sum_{t=2}^T \mathbb{I}[S_t = i, S_{t-1} = j]$$

$$\boldsymbol{\theta} = \begin{bmatrix} \theta_{1|1} & \theta_{1|2} & \theta_{1|3} & \theta_{1|4} & \theta_{1|5} \\ \theta_{2|1} & \theta_{2|2} & \theta_{2|3} & \theta_{2|4} & \theta_{2|5} \\ \theta_{3|1} & \theta_{3|2} & \theta_{3|3} & \theta_{3|4} & \theta_{3|5} \\ \theta_{4|1} & \theta_{4|2} & \theta_{4|3} & \theta_{4|4} & \theta_{4|5} \\ \theta_{5|1} & \theta_{5|2} & \theta_{5|3} & \theta_{5|4} & \theta_{5|5} \end{bmatrix}$$

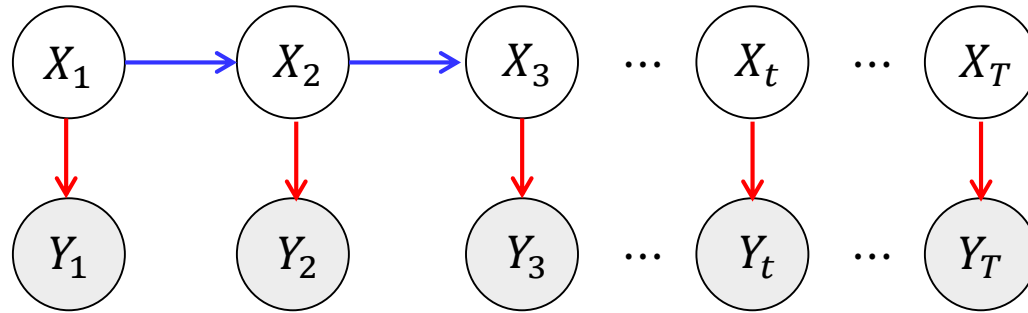
State transition matrix

$$\sum_i \theta_{i|j} = 1$$

If we have $s_{1:t} = \overset{\downarrow}{1}, \overset{\downarrow}{3}, 2, 4, 1, 4, 3, 5, \overset{\downarrow}{1}, \overset{\downarrow}{3}, 4, 2, 1, 4, 4, 2, 4, 5, \overset{\downarrow}{1}, \overset{\downarrow}{3}, 3, 4, \dots \longrightarrow \theta_{3|1} = \frac{3}{5}$

Hidden Markov Model

- The hidden Markov Model (HMM) defines a Markov chain on hidden variables $X_{1:t}$
- The observed variables are dependent on the hidden variables through an emission $P(Y_t|X_t)$



- The joint distribution on the hidden variables and observations are

$$P(X_{1:t}, Y_{1:t}) = P(X_1)P(Y_1|X_1) \prod_{t=2}^T P(X_t|X_{t-1})P(Y_t|X_t)$$

- **Transition distribution:** For a stationary HMM the transition distribution $P(X_t|X_{t-1})$ is defined as the $H \times H$ matrix

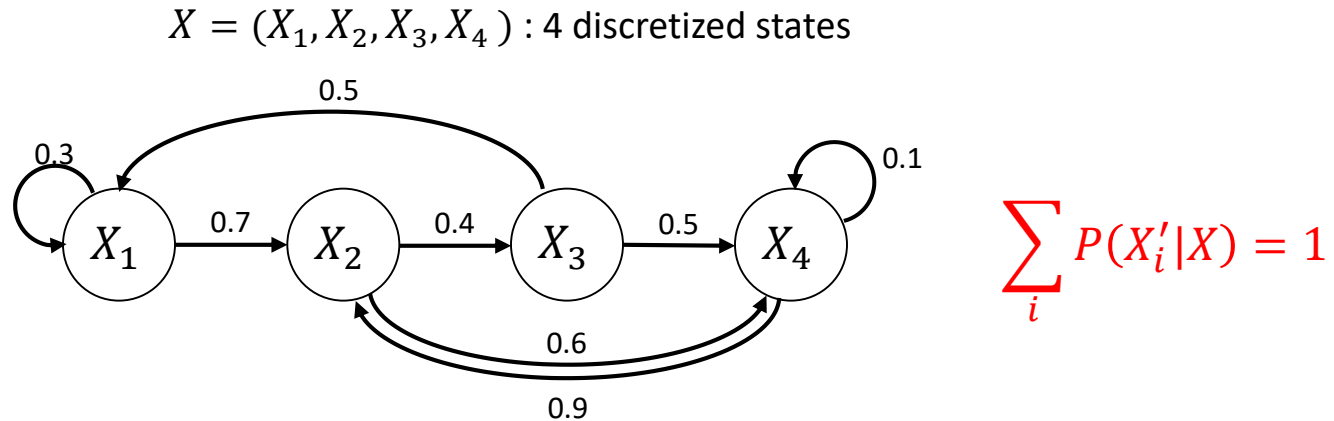
$$M_{i,j} = P(X_t = i | X_{t-1} = j)$$

- **Emission distribution:** For a stationary HMM and emission distribution with discrete states $Y_t \in \{1, \dots, V\}$, we define $V \times H$ matrix

$$O_{i,j} = P(Y_t = i | X_t = j)$$

Hidden Markov Model

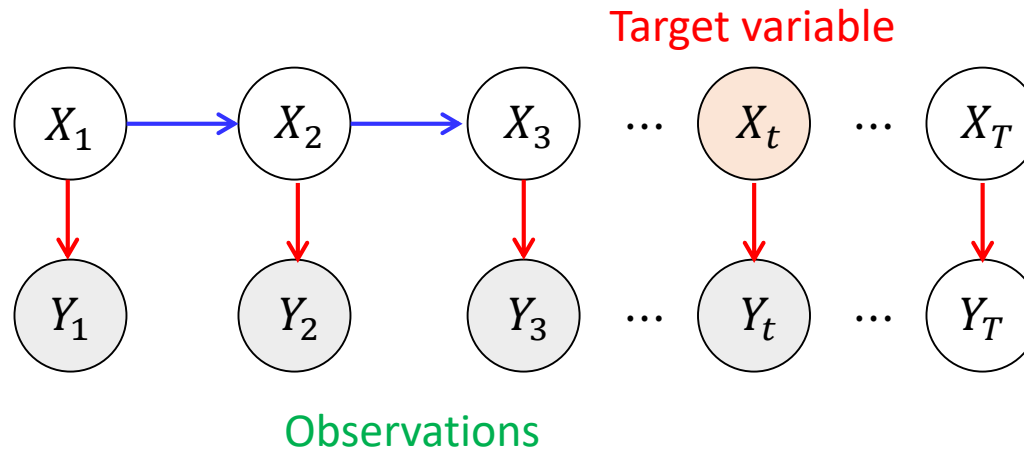
- The state variable X_t is discrete
- The state transition model $P(X'|X)$ is usually sparse,
→ can be represented as a directed graph



- The observation model: $P(Y|X)$ can be deterministic or random

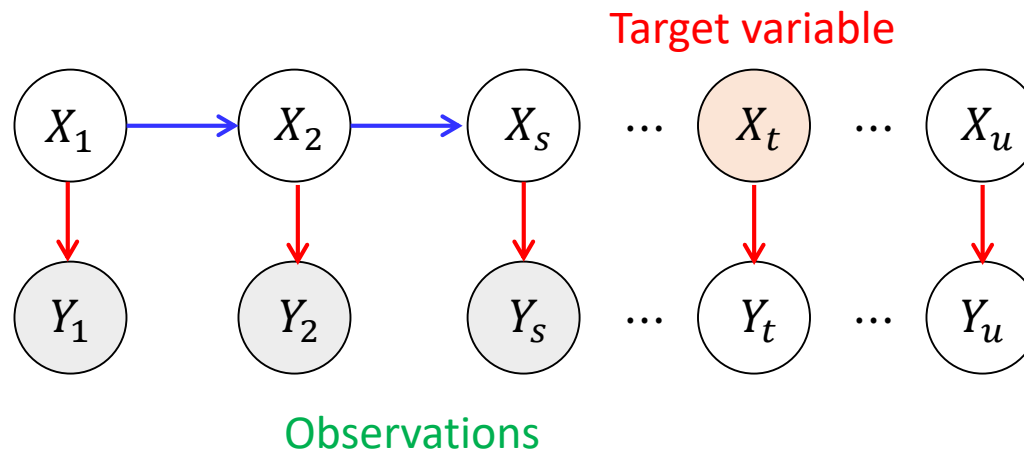
Inferencing of Hidden Markov Model

- Filtering (inferencing the present) $P(x_t|y_{1:t})$
- Prediction (inferencing the future) $P(x_t|y_{1:u}) \quad t > s$
- Smoothing (inferencing the past) $P(x_t|y_{1:u}) \quad t < u$
- Likelihood (inferencing the past) $P(x_{1:t})$
- Most likely hidden path (Viterbi alignment) $\operatorname{argmax}_{x_{1:t}} P(x_{1:t}|y_{1:t})$



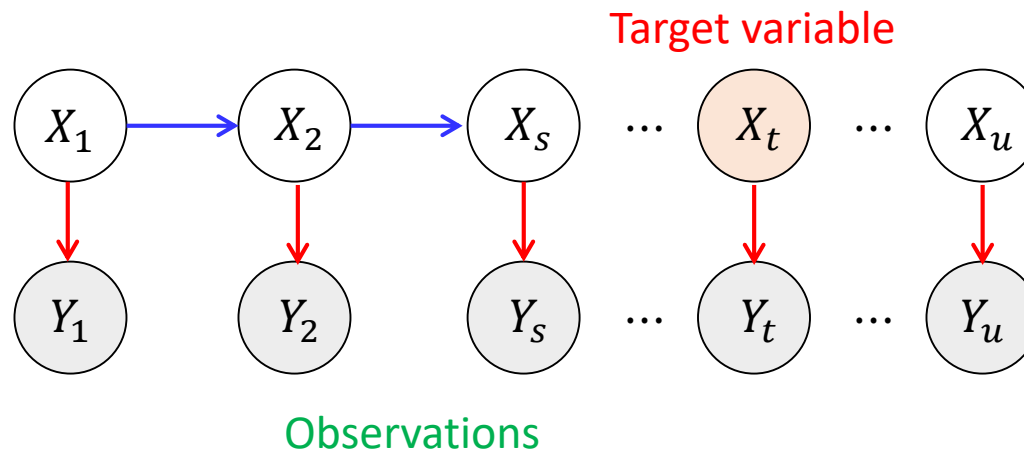
Inferencing of Hidden Markov Model

- Filtering (inferencing the present) $P(x_t|y_{1:t})$
- Prediction (inferencing the future) $P(x_t|y_{1:u}) \ t > s$
- Smoothing (inferencing the past) $P(x_t|y_{1:u}) \ t < u$
- Likelihood (inferencing the past) $P(x_{1:t})$
- Most likely hidden path (Viterbi alignment) $\operatorname{argmax}_{x_{1:t}} P(x_{1:t}|y_{1:t})$



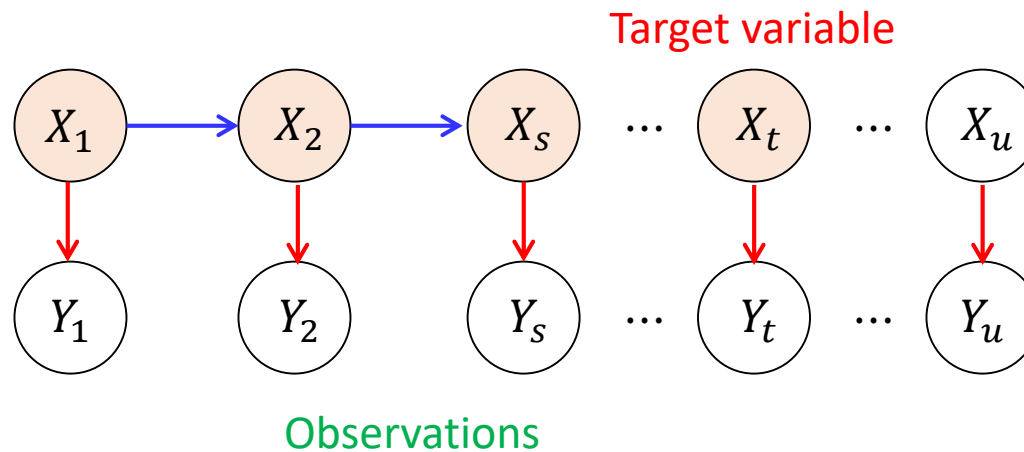
Inferencing of Hidden Markov Model

- Filtering (inferencing the present) $P(x_t|y_{1:t})$
- Prediction (inferencing the future) $P(x_t|y_{1:u}) \ t > s$
- Smoothing (inferencing the past) $P(x_t|y_{1:u}) \ t < u$
- Likelihood (inferencing the past) $P(x_{1:t})$
- Most likely hidden path (Viterbi alignment) $\operatorname{argmax}_{x_{1:t}} P(x_{1:t}|y_{1:t})$



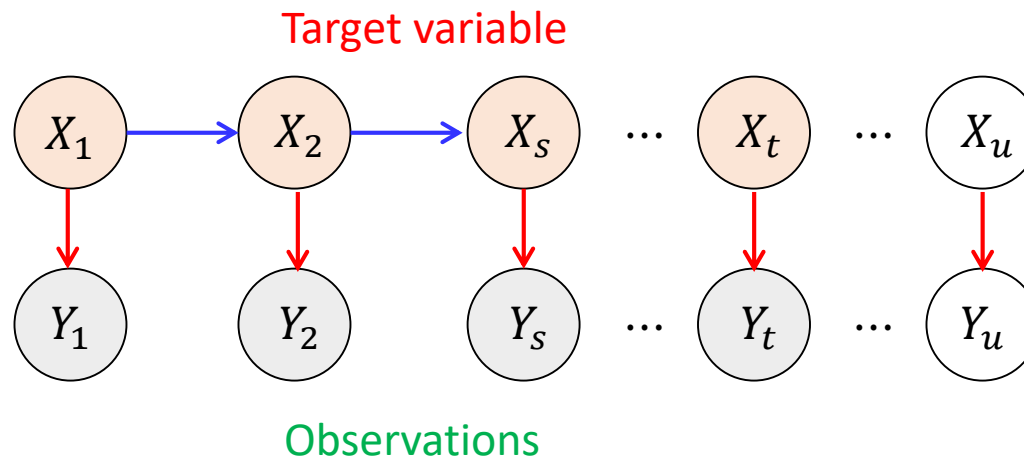
Inferencing of Hidden Markov Model

- Filtering (inferencing the present) $P(x_t|y_{1:t})$
- Prediction (inferencing the future) $P(x_t|y_{1:u}) \ t > s$
- Smoothing (inferencing the past) $P(x_t|y_{1:u}) \ t < u$
- Likelihood (inferencing the past) $P(x_{1:t})$
- Most likely hidden path (Viterbi alignment) $\operatorname{argmax}_{x_{1:t}} P(x_{1:t}|y_{1:t})$



Inferencing of Hidden Markov Model

- Filtering (inferencing the present) $P(x_t|y_{1:t})$
- Prediction (inferencing the future) $P(x_t|y_{1:u}) \quad t > s$
- Smoothing (inferencing the past) $P(x_t|y_{1:u}) \quad t < u$
- Likelihood (inferencing the past) $P(x_{1:t})$
- Most likely hidden path (Viterbi alignment) $\operatorname{argmax}_{x_{1:t}} P(x_{1:t}|y_{1:t})$



Inferencing of Hidden Markov Model

- Filtering (inferencing the present)

$$P(x_t | y_{1:t})$$

$$P(x_t | y_{1:t}) = \frac{P(x_t, y_{1:t})}{P(y_{1:t})} \propto P(x_t, y_{1:t})$$

$$P(x_t, y_{1:t}) = \sum_{x_{t-1}} P(x_t, x_{t-1}, y_{1:t-1}, y_t)$$

$$\sum_{x_{t-1}} P(y_t | y_{1:t-1}, x_t, x_{t-1}) P(x_t | y_{1:t-1}, x_{t-1}) P(x_{t-1}, y_{1:t-1})$$

$$\sum_{x_{t-1}} P(y_t | x_t) P(x_t | x_{t-1}) P(x_{t-1}, y_{1:t-1})$$

∴ Conditional independence

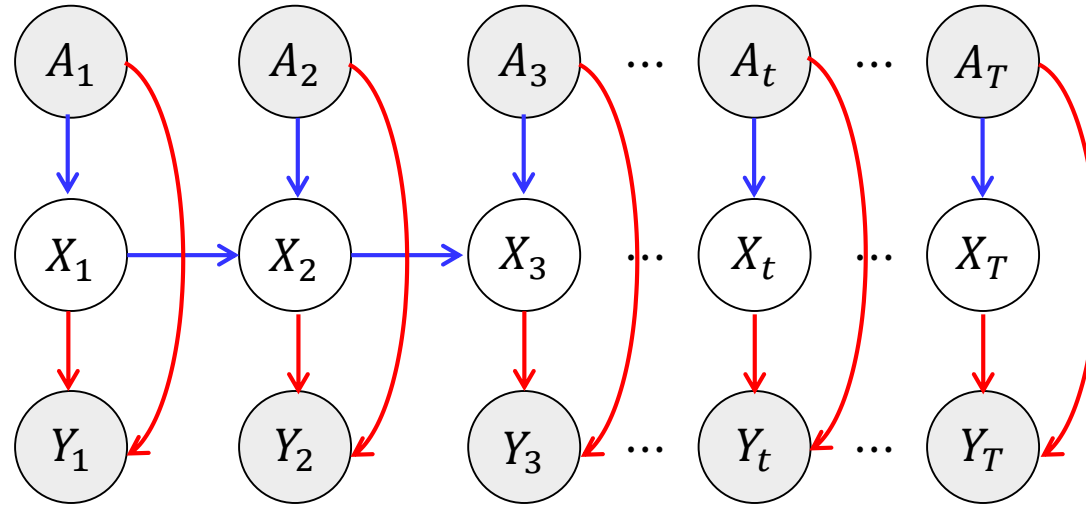
$$\underbrace{P(y_t | x_t)}_{\text{corrector}} \sum_{x_{t-1}} \underbrace{P(x_t | x_{t-1}) P(x_{t-1}, y_{1:t-1})}_{\text{predictor}}$$

- Bayesian view

$$P(x_t | y_{1:t}) \propto \sum_{x_{t-1}} \underbrace{P(y_t | x_t) P(x_t | x_{t-1})}_{\text{Regarded as likelihood}} \underbrace{P(x_{t-1} | y_{1:t-1})}_{\text{Modified prior distribution has an effect of removing all nodes in the graph before time } t-1}$$

Modified prior distribution has an effect of removing all nodes in the graph before time $t - 1$

Input and output hidden Markov Model (IOHMM)



The state transition model : $P(X_t | X_{t-1}, A_t)$

The observation model : $P(Y_t | X_t, A_t)$

Continuous Dynamic Bayesian Network

Continuous-state Markov model

- In many practical time series applications, the data is naturally continuous (i.e., variables are not discretized), particularly for models of the physical environment
- Restrict the form of the continuous transition $p(X_t|X_{t-1})$
- A simple yet powerful class of such transitions are the **linear dynamical systems**
- A **deterministic linear dynamical system** defines the temporal evolution of a vector x_t according to the discrete-time update equation

$$x_t = \mathbf{A}_t x_{t-1}$$

where $\mathbf{A}_t$ is the transition matrix at time t

- If $\mathbf{A}_t$ is invariant with t , the process is called stationary or time-invariant.

Observed linear dynamic system

- A stochastic linear dynamical system defines the temporal evolution of a vector x_t according to the discrete-time update equation

$$x_t = \mathbf{A}_t x_{t-1} + \eta_t$$

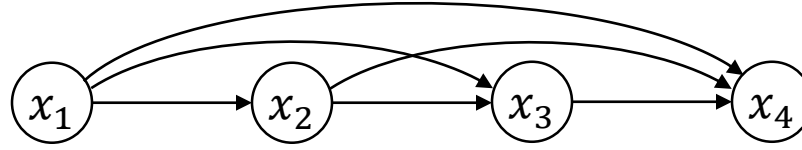
where η_t is a noise vector sampled from a Gaussian distribution

$$\eta_t \sim \mathcal{N}(\mu_t, \Sigma_t)$$

- This is equivalent to a first-order Markov model with transition

$$p(x_t | x_{t-1}) = \mathcal{N}(x_t | \mathbf{A}_t x_{t-1} + \eta_t, \Sigma_t)$$

Auto-Regressive model



- A scalar time-invariant Auto-Regressive (AR) model is defined by

$$x_t = \sum_{l=1}^L a_l x_{t-l} + \eta_t, \quad \eta_t \sim \mathcal{N}(\mu, \sigma^2)$$

where $a = (a_1, a_2, \dots, a_L)^T$ are AR coefficients and σ^2 is innovation noise.

- As a belief network, the AR model can be written as an L th-order Markov model:

$$p(x_{1:T}) = \prod_{t=1}^T p(x_t | x_{t-1}, \dots, x_{t-L}), \quad \text{with } x_i = 0 \text{ for } i \leq 0$$

$\hat{x}_{t-1} = (x_{t-1}, \dots, x_{t-L})$

with $p(x_t | x_{t-1}, \dots, x_{t-L}) = \mathcal{N}(x_t | \sum_{l=1}^L a_l x_{t-l}, \sigma^2) = \mathcal{N}(x_t | a^T \hat{x}_{t-1}, \sigma^2)$

Similar to Bayesian Regression

- Heavily used in financial time series prediction, being able to capture simple trends in the data
- The AR coefficients form a compressed representation of the signal

Training Auto-Regressive model

- Maximum likelihood training of the AR coefficients is straightforward based on

$$\begin{aligned}\log p(x_{1:T}) &= \log \prod_{t=1}^T p(x_t | x_{t-1}, \dots, x_{t-L}) \\ &= \sum_{t=1}^T \log(x_t | \hat{x}_{t-1}) \\ &= -\frac{1}{2\sigma^2} \sum_{t=1}^T (x_t - a^T \hat{x}_{t-1})^2 - \frac{T}{2} \log(2\pi\sigma^2)\end{aligned}$$

- Differentiating with respect to a and equating to zero we arrive at

$$\sum_{t=1}^T (\mathbf{x}_t - a^T \hat{x}_{t-1}) \hat{x}_{t-1} = 0 \quad \rightarrow \quad a = \left[\sum_t \hat{x}_{t-1} \hat{x}_{t-1}^T \right]^{-1} \sum_t \mathbf{x}_t \hat{x}_{t-1}$$

- Similarly, $\mathbf{x}_t$: target output (scalar)

$$\sigma^2 = \frac{1}{T} \sum_{t=1}^T (x_t - a^T \hat{x}_{t-1})^2$$

Time-varying Auto-Regressive model

- Learning the AR coefficients as a problem in inference in a latent linear dynamical system (LDS):

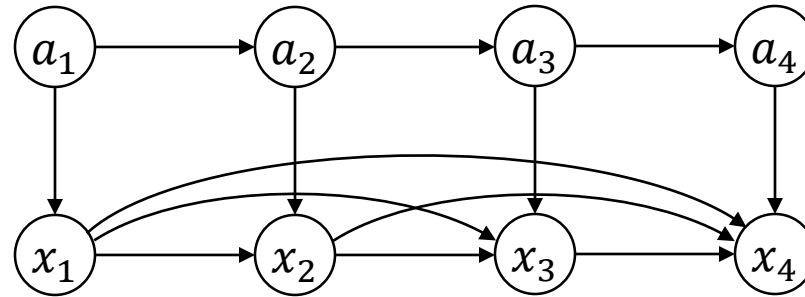
$$x_t = \hat{x}_{t-1}^T a_t + \eta_t, \quad \eta_t \sim \mathcal{N}(0, \sigma^2)$$

which can be viewed as the emission distribution of a latent LDS in which the hidden variable is a_t and the time dependent emission matrix is given by $\hat{x}_{t-1}^T$.

- By placing a simple latent transition

$$a_t = a_{t-1} + \eta_t^a, \quad \eta_t^a \sim \mathcal{N}(0, \sigma_a^2 \mathbf{I})$$

which encourages the AR coefficients to change slowly with time



Time-varying Auto-Regressive model

- Learning the AR coefficients as a problem in inference in a latent linear dynamical system (LDS):

$$x_t = \hat{x}_{t-1}^T a_t + \eta_t, \quad \eta_t \sim \mathcal{N}(0, \sigma^2)$$

which can be viewed as the emission distribution of a latent LDS in which the hidden variable is a_t and the time dependent emission matrix is given by $\hat{x}_{t-1}^T$.

- By placing a simple latent transition

$$a_t = a_{t-1} + \eta_t^a, \quad \eta_t^a \sim \mathcal{N}(0, \sigma_a^2 \mathbf{I})$$

which encourages the AR coefficients to change slowly with time

- The joint distribution between the observation $x_{1:T}$ and the coefficients $a_{1:t}$

$$p(a_{1:T} | x_{1:T}) \propto p(x_{1:T}, a_{1:T}) = \prod_{t=2}^T p(x_t | a_t, \hat{x}_{t-1}) p(a_t | a_{t-1})$$

then we can compute

$$a_{1:T}^* = \underset{a_{1:T}}{\operatorname{argmax}} p(a_{1:T} | x_{1:T})$$

from which **the MAP estimates for the AR coefficients can be determined**

Time-varying variance Auto-Regressive model

- For some applications, particularly in finance, the variance can change with time due to volatility

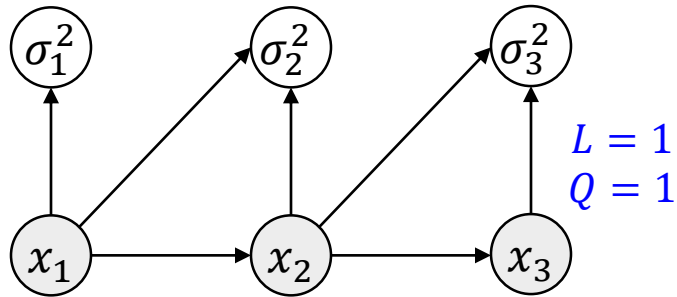
$$x_t = \sum_{l=1}^L a_l x_{t-l} + \eta_t, \quad \eta_t \sim N(0, \sigma_t^2)$$

Time varying variance

$$\bar{x}_t = \sum_{l=1}^L a_l x_{t-l}$$

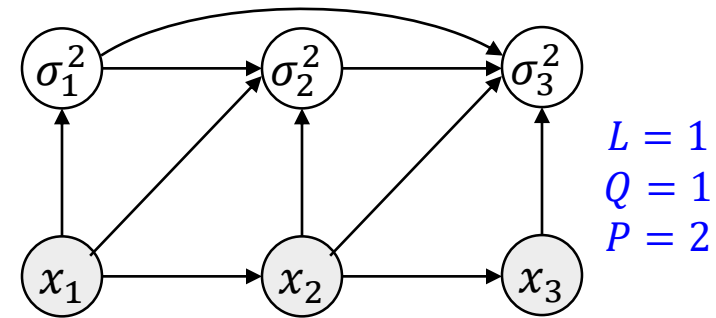
- The estimated time varying variance of noise can be computed

Auto Regressive Conditional Heteroscedasticity (ARCH)



$$\sigma_t^2 = \sigma_0 + \sum_{i=1}^Q \alpha_i (x_{t-i} - \bar{x}_{t-i})^2$$

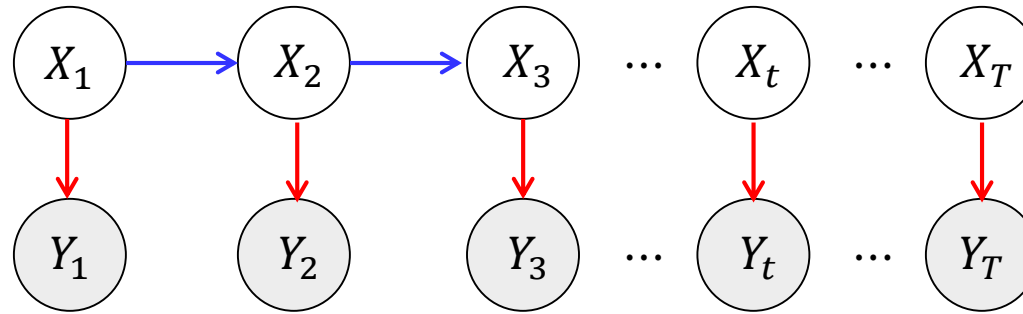
Generalized ARCH model (GARCH)



$$\sigma_t^2 = \sigma_0 + \sum_{i=1}^Q \alpha_i (x_{t-i} - \bar{x}_{t-i})^2 + \sum_{i=1}^P \beta_i \sigma_{t-i}^2$$

Linear Gaussian state space model

- The latent LDS defines a stochastic linear dynamical system in a latent space on a sequence of states $x_{1:T}$
- Observations $y_{1:T}$ are used to infer the hidden states that tracks or explains the system evolution



Transition model : $x_t = \mathbf{A}_t x_{t-1} + \eta_t^x$,

Emission model : $y_t = \mathbf{B}_t x_t + \eta_t^y$,

$\eta_t^x \sim \mathcal{N}(\bar{x}_t, \Sigma_t^x)$

$\eta_t^y \sim \mathcal{N}(\bar{y}_t, \Sigma_t^y)$

$\mathbf{A}_t$: transition matrix

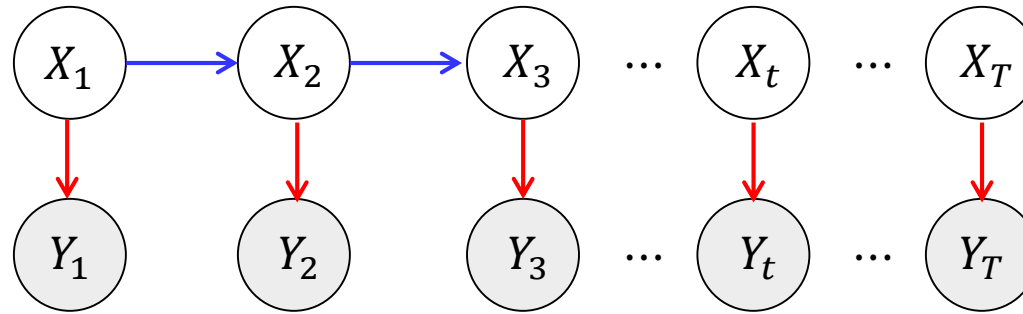
$\mathbf{B}_t$: emission matrix

η_t^x transition noise vector with a hidden bias $\bar{x}_t$

η_t^y emission noise vector with a hidden bias $\bar{y}_t$

Linear Gaussian state space model

- The latent LDS defines a stochastic linear dynamical system in a latent space on a sequence of states $x_{1:T}$
- Observations $y_{1:T}$ are used to infer the hidden states that tracks or explains the system evolution



Transition model : $p(x_t|x_{t-1}) = \text{N}(x_t|\mathbf{A}_t x_{t-1} + \bar{x}_t, \Sigma_t^x)$, $p(x_1) = \text{N}(x_1|\mu_\pi, \Sigma_\pi)$

Emission model : $p(y_t|x_t) = \text{N}(y_t|\mathbf{B}_t x_t + \bar{y}_t, \Sigma_t^y)$

- The first order Markov model is then defined as

$$p(x_{1:T}, y_{1:T}) = p(x_1)p(y_1|x_1) \prod_{t=2}^T p(x_t|x_{t-1})p(y_t|x_t)$$

Kalman filter

- Recall the filtering recursion for HMM:

$$P(x_t|y_{1:t}) \propto \sum_{x_{t-1}} P(y_t|x_t)P(x_t|x_{t-1})P(x_{t-1}|y_{1:t-1})$$

- For linear Gaussian state-space model, the recursion becomes

$$P(x_t|y_{1:t}) \propto \int_{x_{t-1}} P(y_t|x_t)P(x_t|x_{t-1})P(x_{t-1}|y_{1:t-1}), \quad \text{for } t > 1$$

- Since the product of two Gaussians is another Gaussian, and the integral of a Gaussian is another Gaussian, $P(x_t|y_{1:t})$ is Gaussian:

$$P(x_t|y_{1:t}) = N(x_t|f_t, F_t)$$

- Thus, the recursion is for computing the mean f_t and the variance F_t for $P(x_t|y_{1:t})$ using f_{t-1} and the variance F_{t-1} for $P(x_{t-1}|y_{1:t-1})$



$$P(x_{t-1}|y_{1:t-1}) = N(x_{t-1}|f_{t-1}, F_{t-1})$$

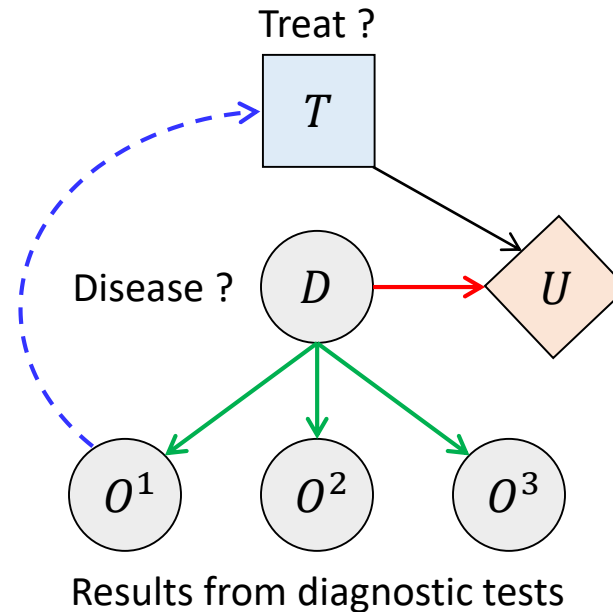
$$P(x_t|y_{1:t}) = N(x_t|f_t, F_t)$$

Influential Diagram

Motivation

Bayesian Network + Decision node + Utility node = Decision network (Influential Diagram)

- make rational decisions based on a probabilistic model and utility function



-  **A chance node** corresponds to a random variable
-  **A decision node** corresponds to each decision to be made
-  **A utility node** corresponds to an additive utility component

Utility theory

- How to compare the plausibility of different statements?

A : “we can be a millinery if we go to graduate school”

vs

B : “we can be a millenary if we go to Samsung”

- If you believe A more than B , you can write $A \succ B$
- If you believe B more than A , you can write $A \prec B$
- If you have the same belief, you can write $A \sim B$

Constraints on Rational Preference

- Assumptions about relationships of $\succ$ and $\sim$
 - Completeness (comparability): either $A \succ B$, $A \prec B$ or $A \sim B$
 - Transitivity: if $A \succ B$ and $B \succ C$, then $A \succ C$
 - Continuity: if $A \succ B \succ C$, there exists a probability p such that $[A: p; C: 1 - p] \sim B$
 - Monotonicity: if $A \succ B$, then for any C and probability p , $[A: p; C: 1 - p] \succ [B: p; C: 1 - p]$
- The degree of belief can be represented by a real-valued function:
 - $P(A) > P(B)$ if and only if $A \succ B$
 - $P(A) = P(B)$ if and only if $A \sim B$

Utility functions

- Just as beliefs can be subjective, so can preferences.
- Preference operators can be used to compare preferences over uncertain outcomes. That is, a lottery is a set of probabilities $p_{1:n}$ associated with a set of outcomes $S_{1:n}$.

$$[S_1:p_1; S_2:p_2; \dots; S_n:p_n]$$

- The utility of a lottery is given by

$$U([S_1:p_1; S_2:p_2; \dots; S_n:p_n]) = \sum_{i=1}^n p_i U(S_i)$$

Maximum expected utility principle

- Reach rational decisions with imperfect knowledge of the state of the world.
- Expected utility of taking action a given that we observe o and take action a :

Probabilistic model Utility function

$P(s'|o, a)$ $U(s')$

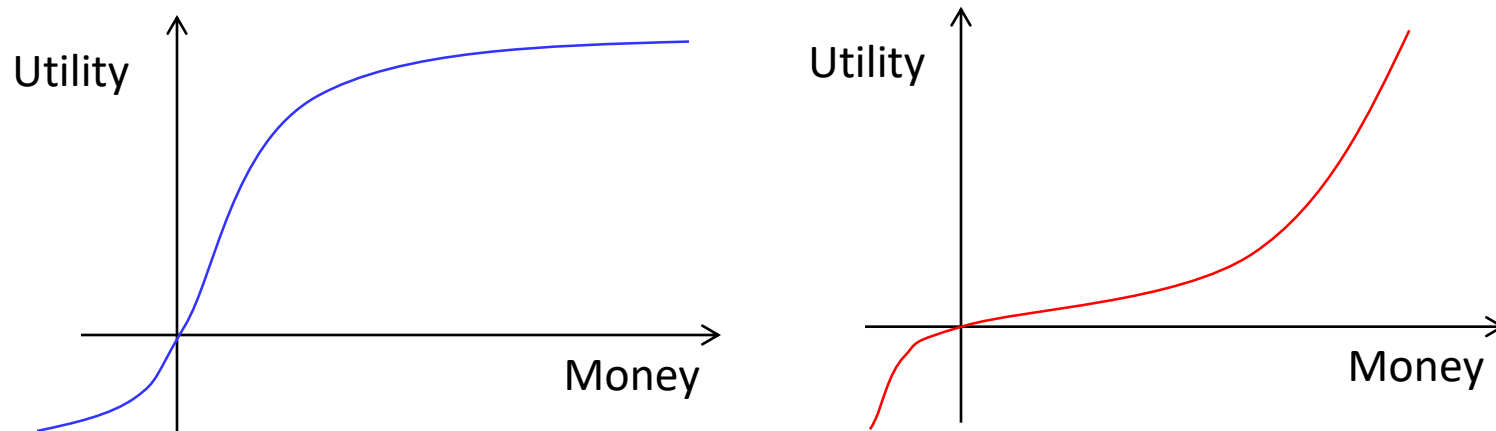
$$EU(a|o) = \sum_{s'} P(s'|o, a) U(s')$$

- The principle of maximum expected utility: a rational agent should choose the action that maximizes expected utility:

$$a^* = \operatorname{argmax}_a EU(a|o)$$

Utility of money

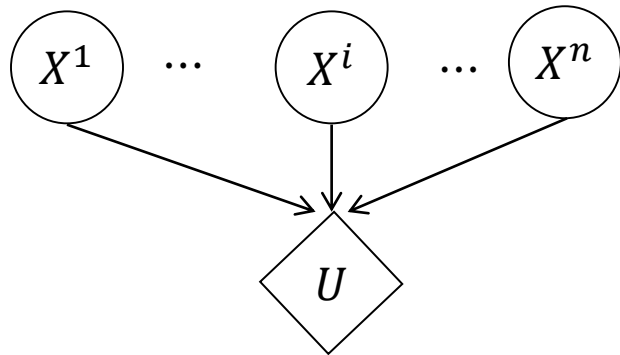
- Monetary values are often used to infer utility function. For example, cost of property and human loss damage caused by natural disasters
- It is well known that the relationship between utility and money is not linear as shown below



A: Wining 1\$ with a probability 1 vs **B:** Wining 100\$ with a probability 0.01

- **Risk averse** : Preference for A (Utility function is concave)
- **Risk neutral** : There is no difference between A and B (Utility function is a linear)
- **Risk seeking** : Preference for B (Utility function is convex)

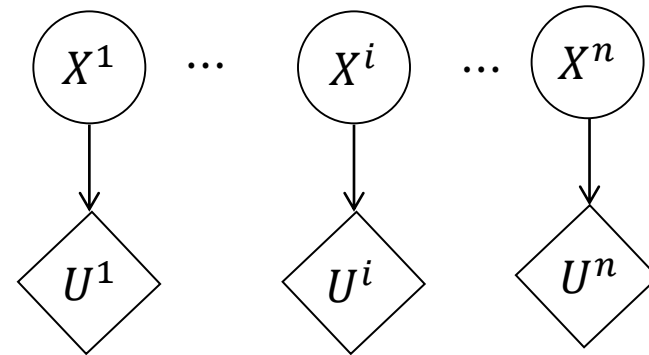
Multiple variable utility function



$$U(X^{1:n})$$

If X^i is a binary variable, 2^n parameters are required to specify $U(X^{1:n})$

Assumption:
Sum of single-
variable utility
function



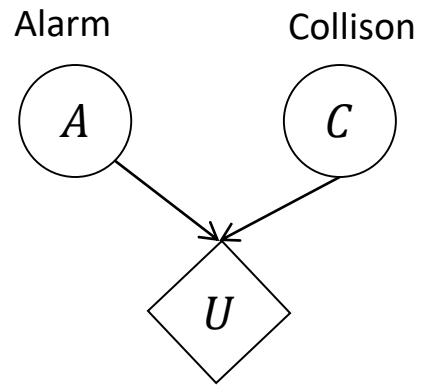
$$U(X^{1:n}) = \sum_{i=1}^n U(X^i)$$

If X^i is a binary variable, $2n$ parameters are required to specify $U(X^{1:n})$

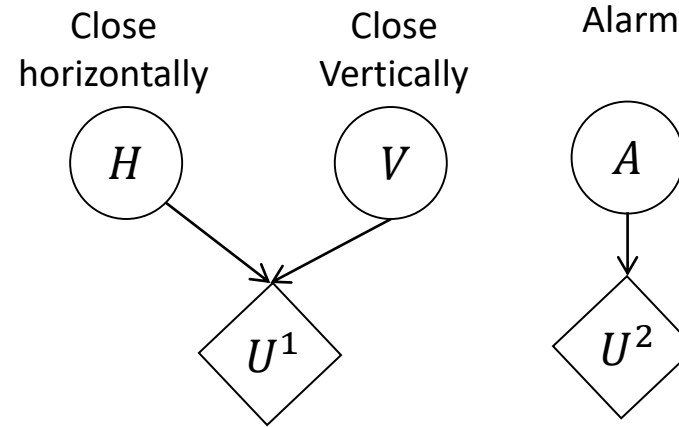
Different additive decomposition can be explicitly imposed on the network structure!

Multiple variable utility function

Example 3.3 (Collision avoidance system)



A	C	U
a^0	c^0	$U(a^0, c^0)$
a^0	c^1	$U(a^0, c^1)$
a^1	c^0	$U(a^1, c^0)$
a^1	c^1	$U(a^1, c^1)$

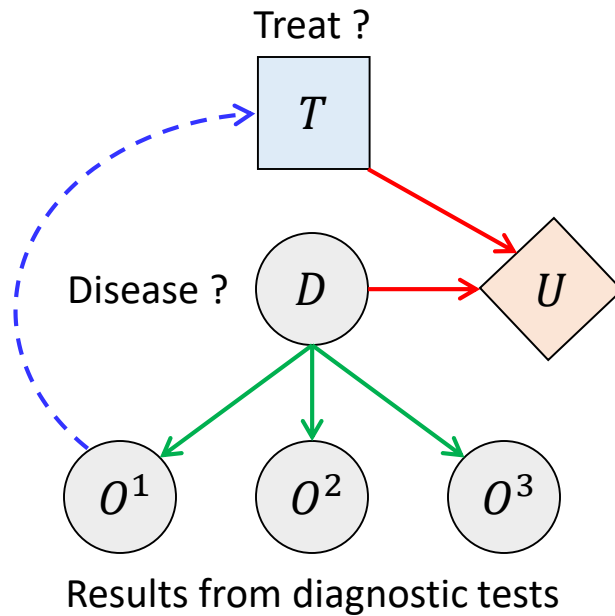


Additive decomposition of
utility function

$$U(h, v, a) = U^1(h, v) + U^2(a)$$

Decision Network

Bayesian Network + Decision node + Utility node = Decision network (Influential Diagram)



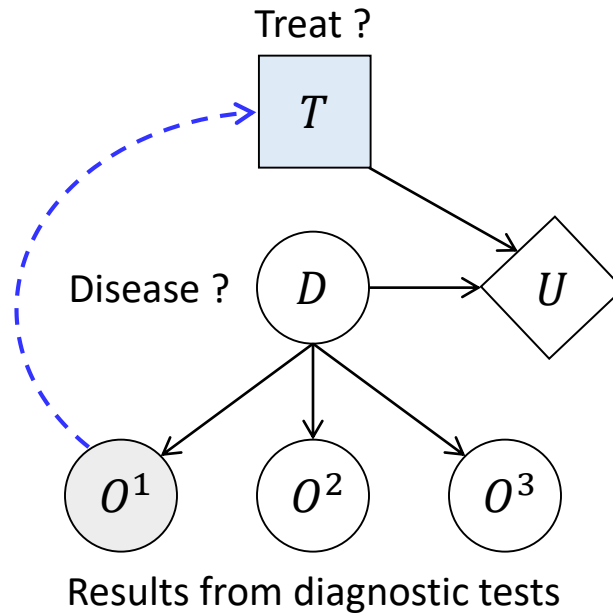
T	D	$U(T, D)$
0	0	0
0	1	-10
1	0	-1
1	1	-1

→ Conditional edge
→ Functional edge
----> Information edge
(often omitted)

- A chance node corresponds to a random variable
- A decision node corresponds to each decision to be made
- ◇ A utility node corresponds to an additive utility component

Decision Network

- Assume we only have a single observation $O^1 = 1 (= o_1^1)$ from test 1



$$\begin{aligned} EU(t^1 | o_1^1) &= \sum_{o_3} \sum_{o_2} \sum_d P(d, o_2, o_3 | t^1, o_1^1) U(t^1, d, o_1^1, o_2, o_3) \\ &= \sum_d \underbrace{P(d | t^1, o_1^1)}_{\text{Can be computed using many inference methods}} U(t^1, d) \end{aligned}$$

Compare $EU(t^0 | o_1^1)$ and $EU(t^1 | o_1^1)$ and chose the treatment that lead maximum EU

Value of information

- It may be beneficial to administer additional diagnostic tests to reduce the uncertainty about the disease. Then, how to choose a test type to be conducted?
- Expected utility of optimal action given observation o :

$$EU^*(o) = \operatorname{argmax}_a EU(a|o)$$

- The value of information (VPI) about new variable O^{new} (**unobserved**) given the current observation o (**observed**):

$$VOI(O^{\text{new}}|o) = \left(\sum_{o^{\text{new}}} P(o^{\text{new}}|o) EU^*(o^{\text{new}}, o) \right) - EU^*(o)$$

- The value of information about a variable is the increase in expected utility with the observation of that variable
- VPI can only capture the increase in expected utility → need to consider the cost associated with observing the new information

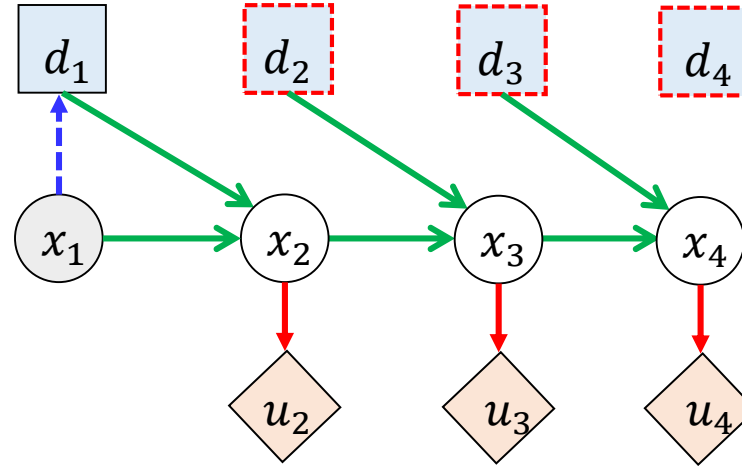
Sequential Decision Making: Partial Ordering

- The sequential decision making problems can be solved by exploiting structure in the problem based on Bayesian Network and the corresponding inference routines
- The sequential decision making problem will be extended later to problems in **control theory** and **reinforcement learning**
- Influential Diagram defines a partial ordering of the nodes:

$$\mathcal{X}_0 < D_1 < \mathcal{X}_1 < D_2, \dots, < \mathcal{X}_{n-1} < D_n < \mathcal{X}_n$$

with $\mathcal{X}_k$ being the variables revealed between decision D_k and D_{k+1} .

Sequential Decision Making: Partial Ordering



- Transition probability :

$$p(x_{t+1}|x_t, d_t)$$

- Utility is

$$U(x_{1:4}) = \sum_{t=2}^4 u(x_t)$$

- The probability of the sequence

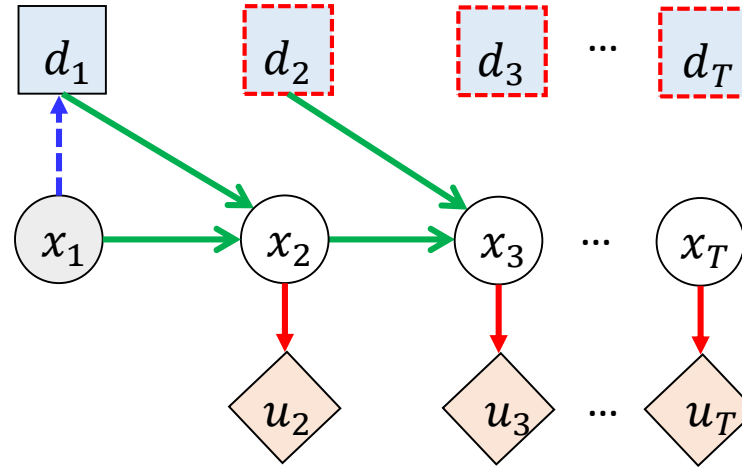
$$p(x_{2:4}|x_1, d_{1:3}) = \prod_{t=1}^3 p(x_{t+1}|x_t, d_t) = p(x_2|x_1, d_1)p(x_3|x_2, d_2)p(x_4|x_3, d_3)$$

- At time $t = 1$, we want to made the decision d_1 that will lead to maximized expected total utility

$$U(d_1|x_1) = \sum_{x_2} \max_{d_2} \sum_{x_3} \max_{d_3} p(x_{2:4}|x_1, d_{1:3}) U(x_{2:4})$$

$$U(d_1|x_1) = \sum_{x_2} \max_{d_2} \sum_{x_3} \max_{d_3} \prod_{t=1}^3 p(x_{t+1}|x_t, d_t) \sum_{t=2}^4 u(x_t)$$

Sequential Decision Making: Partial Ordering



- Transition probability :

$$p(x_{t+1}|x_t, d_t)$$

- Utility is

$$U(x_{1:T}) = \sum_{t=2}^T u(x_t)$$

- The probability of the sequence

$$p(x_{2:T}|x_1, d_{1:T-1}) = \prod_{t=1}^{T-1} p(x_{t+1}|x_t, d_t)$$

- At time $t = 1$, we want to made the decision d_1 that will lead to maximized expected total utility

$$U(d_1|x_1) = \sum_{x_2} \max_{d_2} \sum_{x_3} \max_{d_3} \sum_{x_4} \dots \max_{d_{T-1}} \sum_{x_T} p(x_{2:T}|x_1, d_{1:T-1}) U(x_{2:T})$$

$$U(d_1|x_1) = \sum_{x_2} \max_{d_2} \sum_{x_3} \max_{d_3} \sum_{x_4} \dots \max_{d_{T-1}} \sum_{x_T} \prod_{t=1}^{T-1} p(x_{t+1}|x_t, d_t) \sum_{t=2}^T u(x_t)$$

Sequential Decision Making: Partial Ordering

- The sequential decision making problems can be solved by exploiting structure in the problem based on Bayesian Network and the corresponding inference routines
- The sequential decision making problem will be extended later to problems in **control theory** and **reinforcement learning**
- Influential Diagram defines a partial ordering of the nodes:

$$\mathcal{X}_0 < D_1 < \mathcal{X}_1 < D_2, \dots, < \mathcal{X}_{n-1} < D_n < \mathcal{X}_n$$

with $\mathcal{X}_k$ being the variables revealed between decision D_k and D_{k+1} .

- The optimal first decision $D_1 = d_1$ is determined by computing

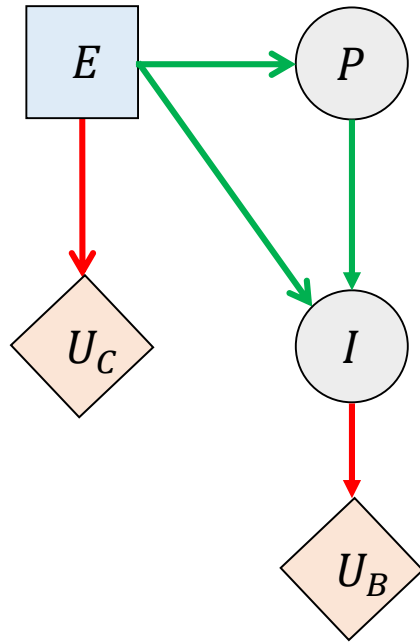
$$U(d_1|x_0) \equiv \sum_{\mathcal{X}_1} \max_{D_2} \dots \sum_{\mathcal{X}_{n-1}} \max_{D_n} \sum_{\mathcal{X}_n} \prod_{i \in \mathcal{L}} p(x_i | \text{pa}(x_i)) \sum_{j \in \mathcal{T}} U_j(\text{pa}(u_j))$$

where $\mathcal{L}$ is a set of random variables and $\mathcal{T}$ is a set of utility variables

- The optimal first decision D_1^* is determined as
$$d_1^* = \operatorname{argmax}_{d_1} U(d_1|x_0)$$

Sequential Decision Making: Partial Ordering

Example 3.4 (Should I do a PhD?)



Do PhD to wind a Nobel Prize?

The ordering : $E^* < \{I, P\}$

Domains

- $\text{dom}(E) = \{\text{do PhD, no PhD}\}$
- $\text{dom}(P) = \{\text{prize, no prize}\}$
- $\text{dom}(I) = \{\text{low, average, high}\}$

Utilities

- $U_C(\text{do PhD}) = -50000$, $U_C(\text{no PhD}) = 0$
- $U_B(\text{low}) = 100000$, $U_B(\text{average}) = 200000$, $U_B(\text{high}) = 500000$

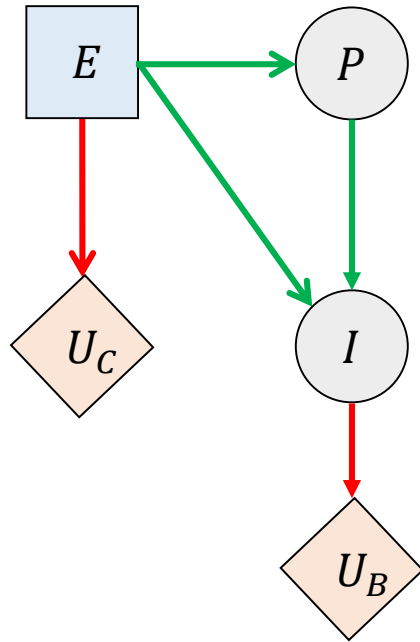
Probabilities

- $p(\text{win Nobel Prize}|\text{no PhD}) = 0.0000001$, $p(\text{win Nobel Prize}|\text{do PhD}) = 0.001$
- $p(\text{low}|\text{do PhD, no prize}) = 0.1$, $p(\text{average}|\text{do PhD, no prize}) = 0.5$, $p(\text{high}|\text{do PhD, no prize}) = 0.4$
- $p(\text{low}|\text{no PhD, no prize}) = 0.2$, $p(\text{average}|\text{no PhD, no prize}) = 0.6$, $p(\text{high}|\text{no PhD, no prize}) = 0.2$
- $p(\text{low}|\text{do PhD, prize}) = 0.01$, $p(\text{average}|\text{do PhD, prize}) = 0.04$, $p(\text{high}|\text{do PhD, prize}) = 0.95$
- $p(\text{low}|\text{no PhD, prize}) = 0.01$, $p(\text{average}|\text{no PhD, prize}) = 0.04$, $p(\text{high}|\text{no PhD, prize}) = 0.95$



Sequential Decision Making: Partial Ordering

Example 3.4 (Should I do a PhD?)



Do PhD to wind a Nobel Prize?

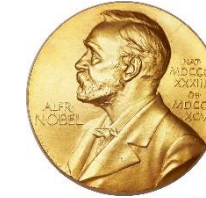
The ordering : $E^* < \{I, P\}$

Domains

- $\text{dom}(E) = \{\text{do PhD, no PhD}\}$
- $\text{dom}(P) = \{\text{prize, no prize}\}$
- $\text{dom}(I) = \{\text{low, average, high}\}$

Utilities

- $U_C(\text{do PhD}) = -50000$, $U_C(\text{no PhD}) = 0$
- $U_B(\text{low}) = 100000$, $U_B(\text{average}) = 200000$, $U_B(\text{high}) = 500000$



Probabilities

- $p(\text{win Nobel Prize}|\text{no PhD}) = 0.0000001$, $p(\text{win Nobel Prize}|\text{do PhD}) = 0.001$
- $p(\text{low}|\text{do PhD, no prize}) = 0.1$, $p(\text{average}|\text{do PhD, no prize}) = 0.5$, $p(\text{high}|\text{do PhD, no prize}) = 0.4$
- $p(\text{low}|\text{no PhD, no prize}) = 0.2$, $p(\text{average}|\text{no PhD, no prize}) = 0.6$, $p(\text{high}|\text{no PhD, no prize}) = 0.2$
- $p(\text{low}|\text{do PhD, prize}) = 0.01$, $p(\text{average}|\text{do PhD, prize}) = 0.04$, $p(\text{high}|\text{do PhD, prize}) = 0.95$
- $p(\text{low}|\text{no PhD, prize}) = 0.01$, $p(\text{average}|\text{no PhD, prize}) = 0.04$, $p(\text{high}|\text{no PhD, prize}) = 0.95$

The expected utility of Education is

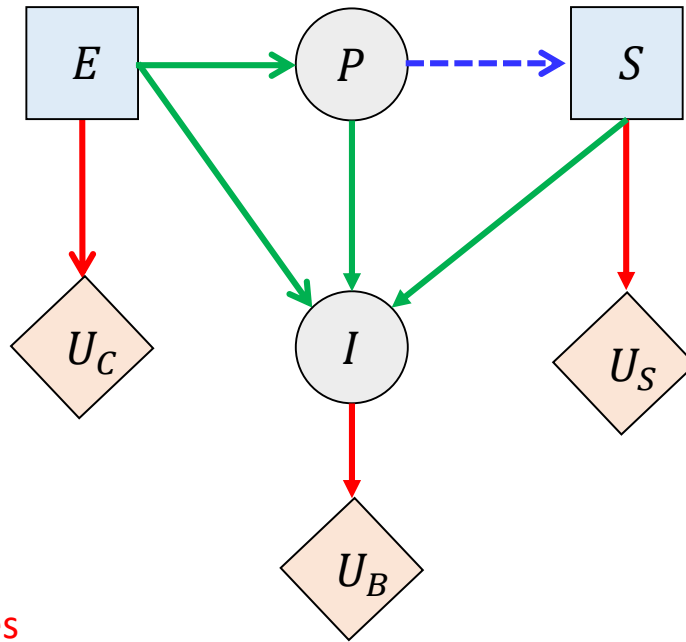
$$U(E) = \sum_{I,P} p(I|E,P)p(P|E)[U_C(E) + U_B(I)]$$

$$U(\text{do PhD}) = 260174$$

$$U(\text{no PhD}) = 240000$$

Sequential Decision Making: Partial Ordering

Example 3.5 (PhD and start-up companies)



Do PhD to wind a Nobel Prize and start-up?

The ordering : $E^* < P < S^* < I$

Domains

- $\text{dom}(E) = \{\text{do PhD, no PhD}\}$
- $\text{dom}(P) = \{\text{prize, no prize}\}$
- $\text{dom}(I) = \{\text{low, average, high}\}$
- $\text{dom}(S) = \{\text{yes, no}\}$

Utilities

- $U_C(\text{do PhD}) = -50000$, $U_C(\text{no PhD}) = 0$
- $U_B(\text{low}) = 100000$, $U_B(\text{average}) = 200000$, $U_B(\text{high}) = 500000$
- $U_S(\text{start up}) = -200000$, $U_S(\text{no start up}) = 0$

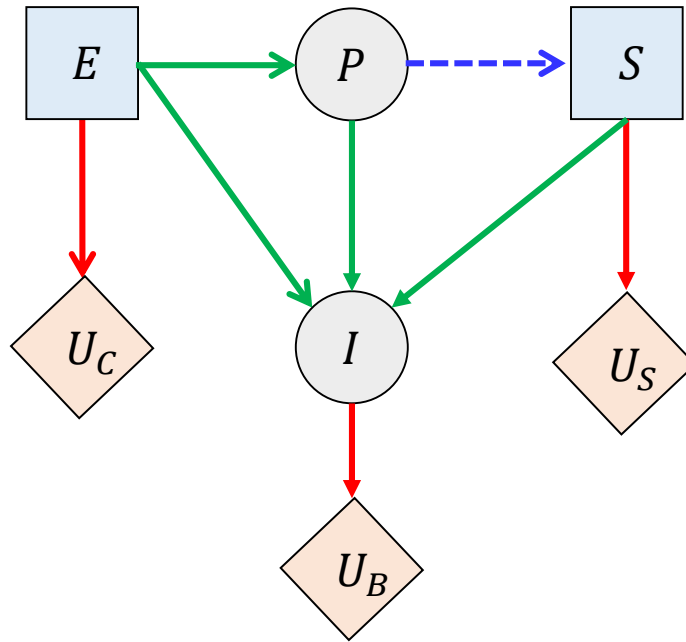
Probabilities

- $p(\text{win Nobel Prize}|\text{no PhD}) = 0.0000001$, $p(\text{win Nobel Prize}|\text{do PhD}) = 0.001$
- $p(\text{low}|\text{do PhD, no prize}) = 0.1$, $p(\text{average}|\text{do PhD, no prize}) = 0.5$, $p(\text{high}|\text{do PhD, no prize}) = 0.4$
- $p(\text{low}|\text{no PhD, no prize}) = 0.2$, $p(\text{average}|\text{no PhD, no prize}) = 0.6$, $p(\text{high}|\text{no PhD, no prize}) = 0.2$
- $p(\text{low}|\text{do PhD, prize}) = 0.01$, $p(\text{average}|\text{do PhD, prize}) = 0.04$, $p(\text{high}|\text{do PhD, prize}) = 0.95$
- $p(\text{low}|\text{no PhD, prize}) = 0.01$, $p(\text{average}|\text{no PhD, prize}) = 0.04$, $p(\text{high}|\text{no PhD, prize}) = 0.95$
- $p(\text{low}|\text{start up, no prize}) = 0.1$, $p(\text{average}|\text{start up, no prize}) = 0.5$, $p(\text{high}|\text{start up, no prize}) = 0.4$
- $p(\text{low}|\text{no start up, no prize}) = 0.2$, $p(\text{average}|\text{no start up, no prize}) = 0.6$, $p(\text{high}|\text{no start up, no prize}) = 0.2$
- $p(\text{low}|\text{start up, prize}) = 0.005$, $p(\text{average}|\text{start up, prize}) = 0.005$, $p(\text{high}|\text{start up, prize}) = 0.99$
- $p(\text{low}|\text{no start up, prize}) = 0.05$, $p(\text{average}|\text{no start up, prize}) = 0.15$, $p(\text{high}|\text{no start up, prize}) = 0.8$



Sequential Decision Making: Partial Ordering

Example 3.5 (PhD and start-up companies)



Do PhD to wind a Nobel Prize and start-up?

The ordering : $E^* < P < S^* < I$

Domains

- $\text{dom}(E) = \{\text{do PhD, no PhD}\}$
- $\text{dom}(P) = \{\text{prize, no prize}\}$
- $\text{dom}(I) = \{\text{low, average, high}\}$
- $\text{dom}(S) = \{\text{yes, no}\}$

Utilities

- $U_C(\text{do PhD}) = -50000$, $U_C(\text{no PhD}) = 0$
- $U_B(\text{low}) = 100000$, $U_B(\text{average}) = 200000$, $U_B(\text{high}) = 500000$
- $U_S(\text{start up}) = -200000$, $U_S(\text{no start up}) = 0$



- Our interest is to advise whether or not it is desirable to take a PhD, bearing mind that later one may or may not win the Nobel Prize, and may or may not form a start-up company
- The expected optimal utility for any state E is

$$U(E) = \sum_P \max_S \sum_I p(I|S,P)p(P|E)[U_C(E) + U_B(I) + U_C(S)]$$

(where we assume that the optimal decisions are taken in the future)

$$U(\text{do PhD}) = 190195$$

$$U(\text{no PhD}) = 240002$$

Questions?

?